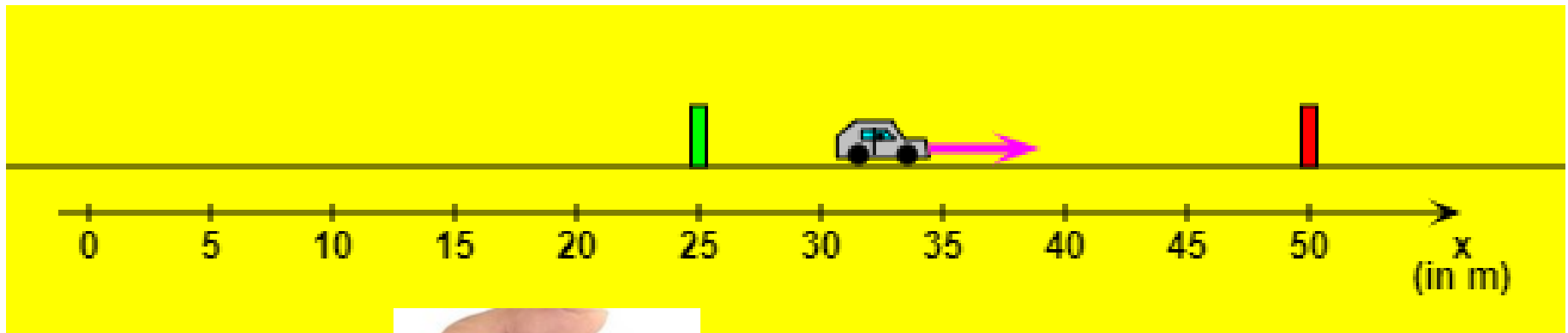
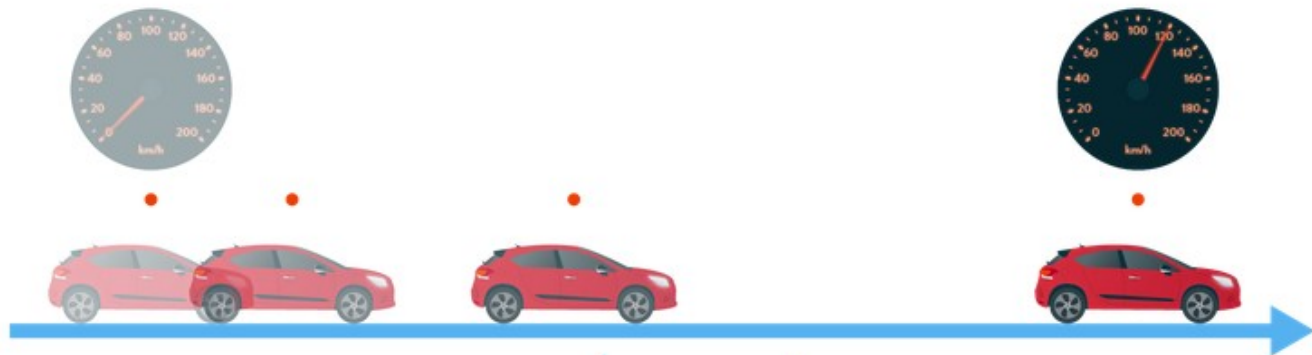
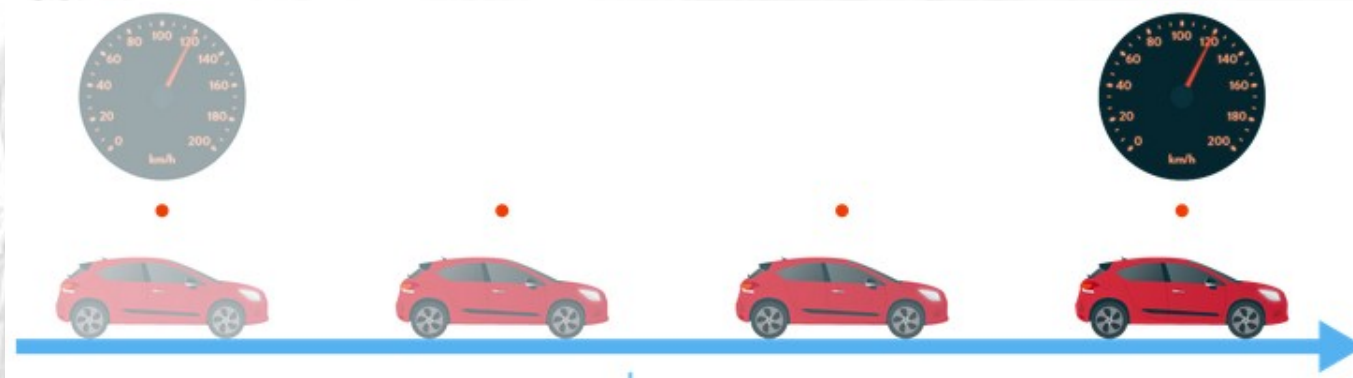


Unit2

Kinematics in One Dimension



Discuss the motions



Kinematics deals with the concepts that are needed to describe motion.

Dynamics deals with the effect that forces have on motion.

Together, kinematics and dynamics form the branch of physics known as ***Mechanics***.

Equation sheet (tests)

equations

$s. = d/t$ average speed

$\Delta x = X_f - X_i$ displacement

$V_{av} = \Delta x/t$ average velocity

$\Delta V = V_f - V_i$ change in velocity

$a. = \Delta V/t$ acceleration

Kinematics equations 1D:

1) position x (location on number line)
 $x = x_i + V_i t + 0.5 a t^2$

Or displacement ($x - x_i$ how far)
 $\Delta x = V_i t + 0.5 a t^2$

2) final velocity (instantaneous) $V = V_i + a t$

3) acceleration $a = (V - V_i)/t$

4) $V^2 = V_i^2 + 2a(x - x_i)$ or $V^2 = V_i^2 + 2a(\Delta x)$

5) Average velocity $= \Delta x/t$ or $(V + V_i)/2$

Physical quantities

t. time elapsed (seconds) – same as Δt

s. average speed (m/s)

d. distance (m) – always positive – how far

X position (m) on number line. Can be < 0

X_f or X final position (m) **X_i** initial position (m)

ΔX (m) displacement = change in position in 1D.
Can be < 0 if moving @ left or up

V_{av} (m/s) average velocity

V (m/s) velocity . Can be < 0 if moving @ left or up

V_i initial velocity **V_f or V** final velocity

a. acceleration (m/s/s) Can be negative if push/pull @ left

2.2 *Speed and Velocity*

Average speed is the distance traveled divided by the time required to cover the distance.

$$\text{Average speed} = \frac{\text{Distance}}{\text{Elapsed time}}$$

SI units for speed: **meters per second** (m/s)

What's your escape speed from this room?
Without me noticing you sneaked out

2.2 *Speed and Velocity*

Example 1 Distance Run by a Jogger

How far does a jogger run in 1.5 hours if his average speed is 2.22 m/s?

$$\text{Average speed} = \frac{\text{Distance}}{\text{Elapsed time}}$$

Question: what's your escape speed from this class?

$$\text{Average speed} = \frac{\text{Distance}}{\text{Elapsed time}}$$

Examples

1) How far does a jogger run in 1.5 hours (convert 1st in s) if his average speed is 2.22 m/s? Convert to km

Convert to miles (1 mile = 1,609m)

2) If it takes you 45 seconds to run 340m, what is your average speed?

3) How long does it take a skier to travel 650m, going 18.5m/s ?

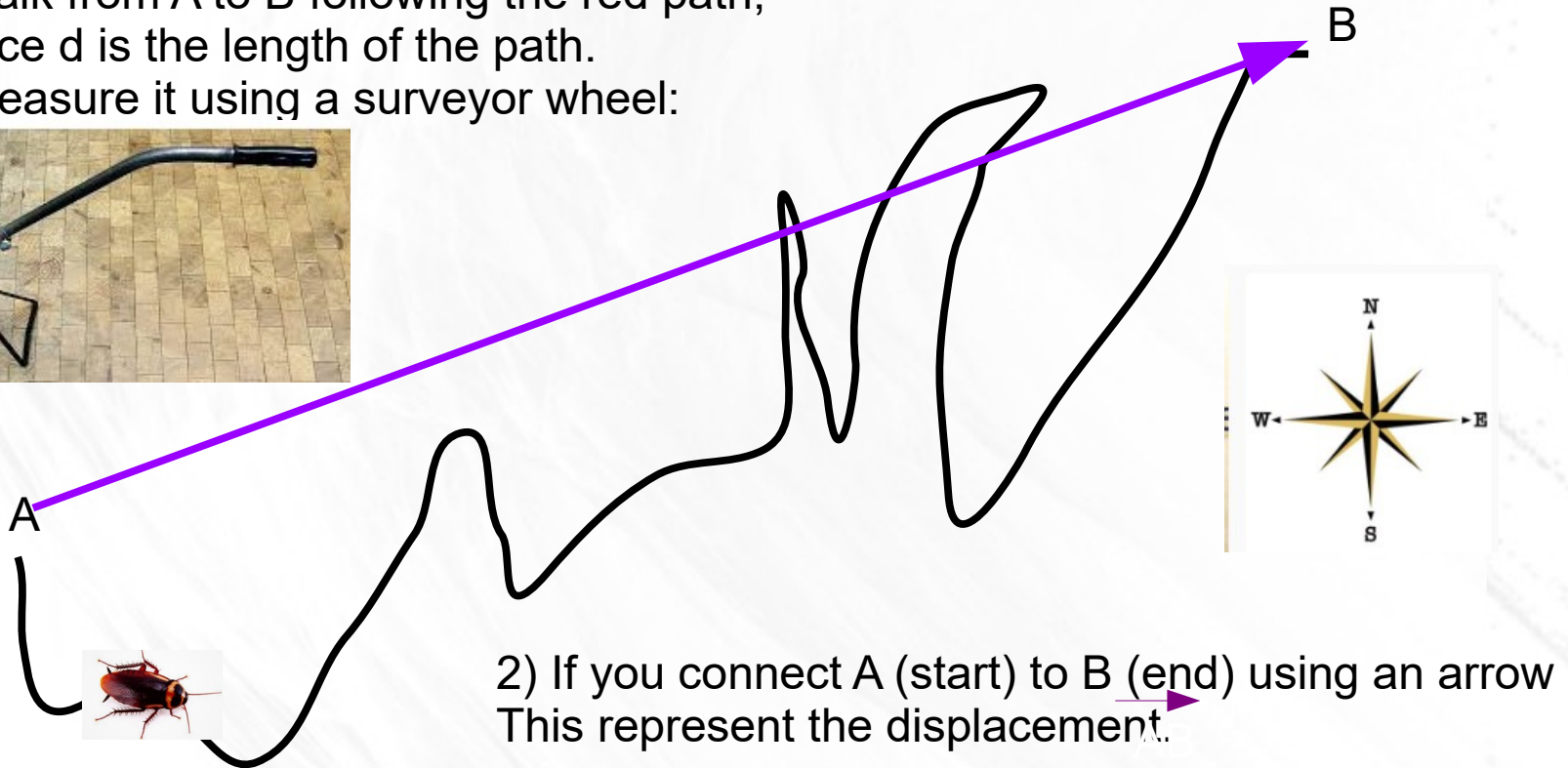
4) at 9:00AM you start on a walk. At 10:30AM you are 4.6 miles from Your starting point. what is your average speed in mph ?

Convert to m/s (Sorry units used in Physics are meters, Seconds, kilograms, joule ..) 1 mile = 1609m

5) What is the average speed of a runner who finishes the 1,500 meters race in 3 minutes 30 seconds ? In m/s

In Physics displacement and distance are not the same physical quantities.

1) If you walk from A to B following the red path, The distance d is the length of the path.
You can measure it using a surveyor wheel:



2) If you connect A (start) to B (end) using an arrow This represent the displacement.

In Physics/Math displacement AB is called a vector

A displacement has a length (length of arrow) and a direction (like 30NW) .

BUT do not worry (for now) in 1D, displacement is easy to understand.
Motion is either @ left (displacement <0) or @ right (displacement >0)

Coordinate System

1. Origin

2. Axis

positive coordinate direction

3. Unit Vector



A toy car starts at $x_0 = 10\text{m}$ ($t = 0$)
Moves at a constant velocity 5m/s
Where is it $t = 3\text{s}$? ($X(t)$)

Kinematics:

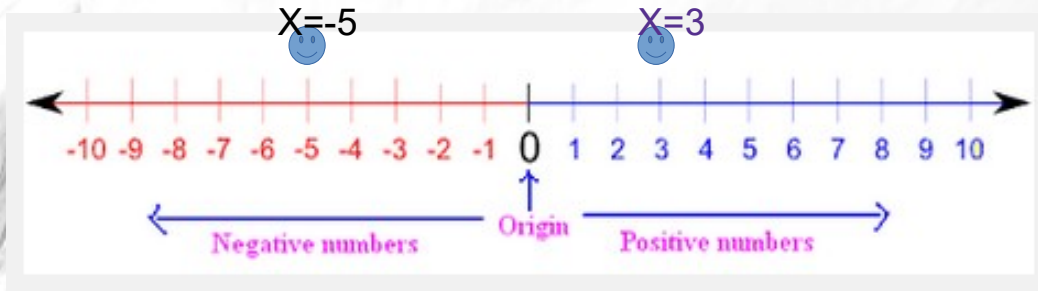
Position as a function of time $x(t)$

Velocity as a function of time $V(t)$

Acceleration as a function of time $a(t)$

Verify with the app

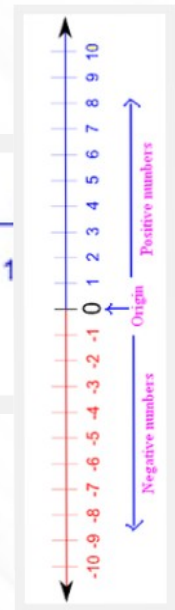
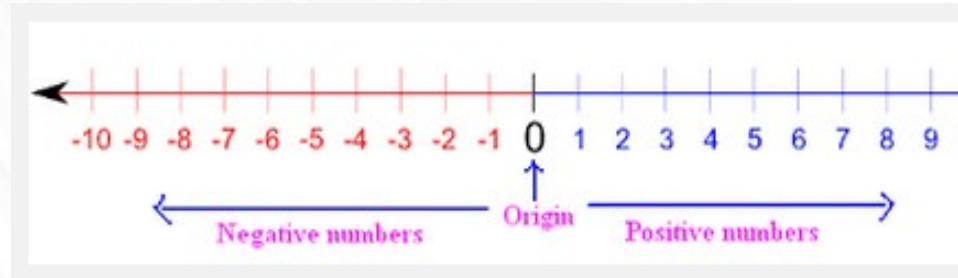
Position x: location on number line



Finding the displacement
In 1D is easy =
Change in position on number line

$$\Delta x = x_2 - x_1 = -5 - 3 = -8 \text{ units}$$

displacement Δx : change in position in 1D
Connect initial position to final position

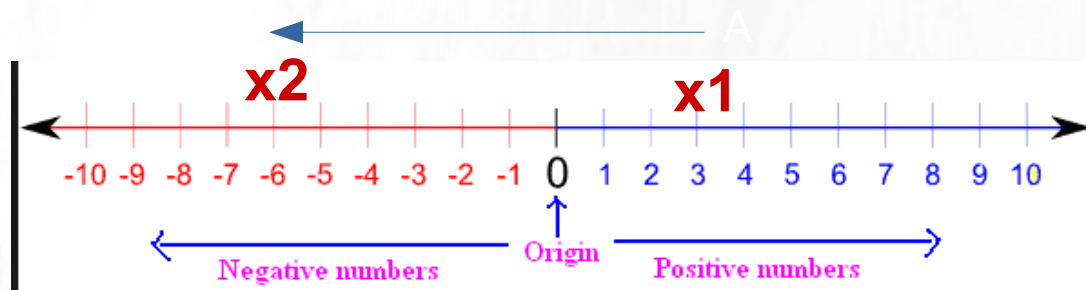


If $X_1 = -5$ and displacement = 7 what is the final position X_2 ?

Same principle can apply to the vertical scale.

If $Y_1 = -3$ and the $Y_2 = 2$ what is the displacement

If the Y_1 is -10 and the displacement is 5 what is the final position Y_2 ?



In one dimension the displacement is just the change of position on the number line.

Displacement = final position – initial position
Displacement is noted Δx

$\Delta x = x_2 - x_1 = x_{\text{final}} - x_{\text{initial}}$

Here the displacement is $x_2 - x_1 = -6 - 3 = -9$ units.

Motion @ left => the displacement is negative

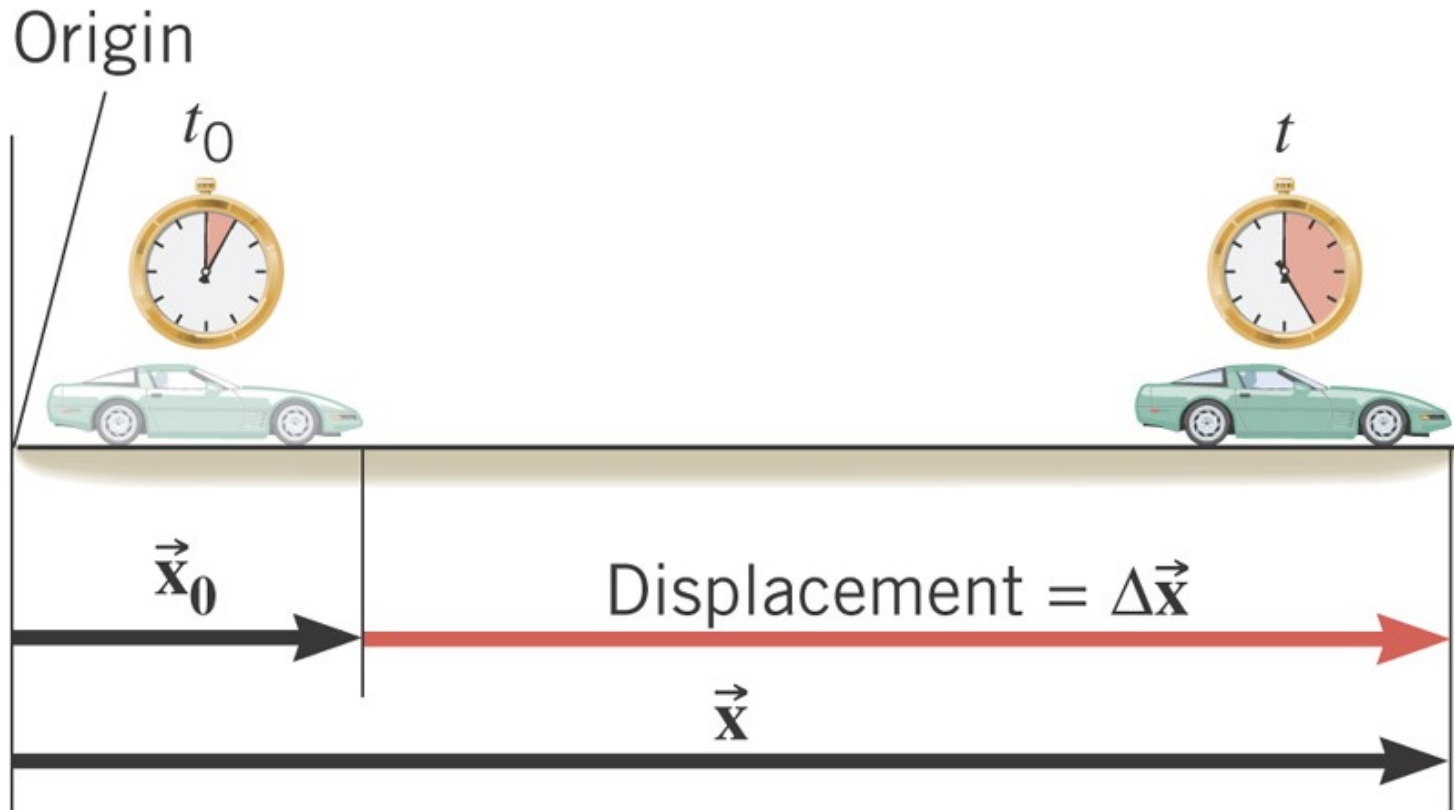
Motion @ right => the displacement is positive

The distance is always positive. Its called a **scalar**. Here the distance is 9 units¹

Displacement in physics is a physical quantity called **vector**.

It has a magnitude, a direction and a unit.

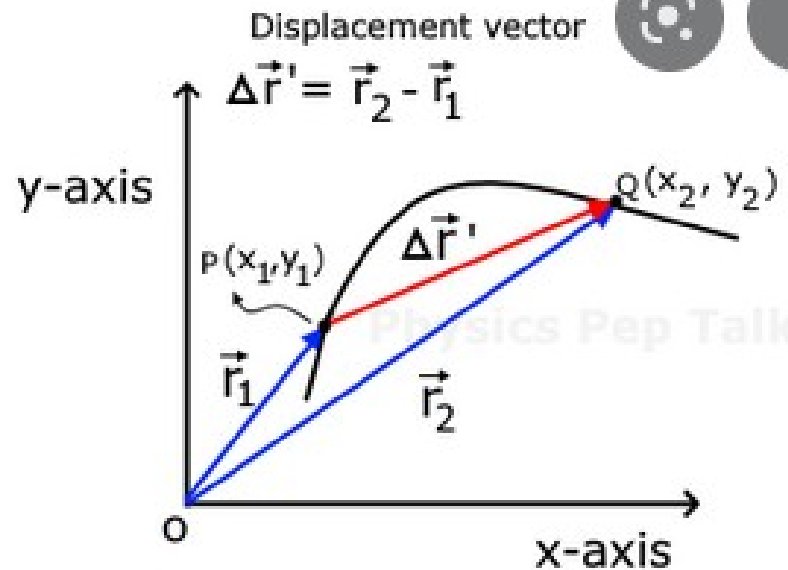
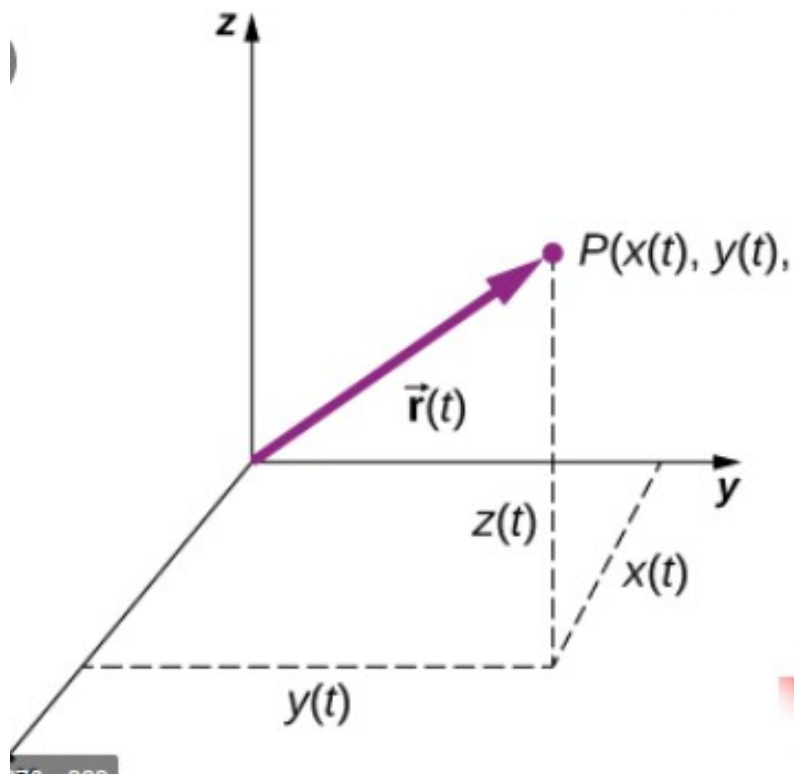
2.1 Displacement



$\vec{x}_o$ = initial position

$\vec{x}$ = final position

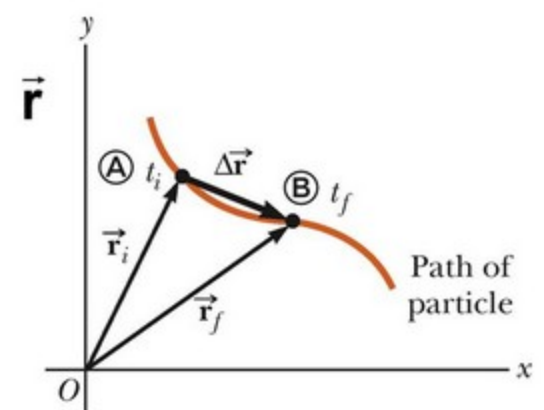
$$\Delta \vec{x} = \vec{x} - \vec{x}_o = \text{displacement}$$



Position and Displacement

- The position of an object is described by its position vector,
- The **displacement** of the object is defined as the **change in its position**

$$\Delta \vec{r} = \vec{r}_f - \vec{r}_i$$



A physical quantity that has a direction and a magnitude like a displacement is called a vector. It is represented by an arrow.

$$\xrightarrow{5} + \xrightarrow{5} = \xrightarrow{10}$$

It is easy to add vectors in 1D. Complete the page.
Draw the vector sum and write its notation.

$$\xrightarrow{5} + \xleftarrow{5} =$$

$$\xrightarrow{5} + \xrightarrow{10} =$$

You walk 5 yards @ right and another 5 yards @ right . The net sum is **10 yards @ right**

$$\xrightarrow{5} + \xleftarrow{10} =$$

10 yards @ left is simply -10 yards.

Since there are only 2 directions. Minus is for left.

$$\xrightarrow{5} + \xleftarrow{15} =$$

Up is positive
Down is negative.

$$\uparrow 10 + \downarrow 5 =$$

$$\uparrow 500 + \downarrow 2000$$

$$\uparrow 500 + \downarrow 1500$$

$$\downarrow 4000 + \uparrow 500$$

2.2 *Speed and Velocity*

Average velocity is the displacement divided by the elapsed time.

$$\text{Average velocity} = \frac{\text{Displacement}}{\text{Elapsed time}}$$

$$\vec{\bar{v}} = \frac{\vec{x} - \vec{x}_o}{t - t_o} = \frac{\Delta \vec{x}}{\Delta t}$$

In Physics Velocity and speed are not the same physical quantity:

average velocity = displacement / time : $V = \Delta x / \text{time}$

If the object is moving @right, velocity is positive

If the object is moving @left, velocity is negative

So to find the velocity : find the displacement (see previous slides) and divide by the time !

velocity is called a vector – notation is $\vec{V}$

It has a magnitude, a direction (which way: left or right if 1D motion), a unit (m/s)

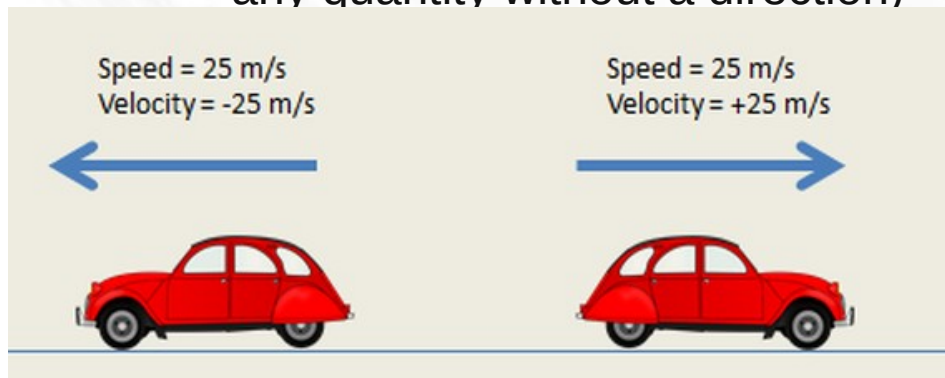
(other examples of vectors: force, acceleration, displacement)

average speed = distance / time = d/t (see previous slides)

speed is not a vector (it is called a scalar) – it has a magnitude, a unit (m/s)

(other examples of scalars: time, energy, distance, temperature, volume ...

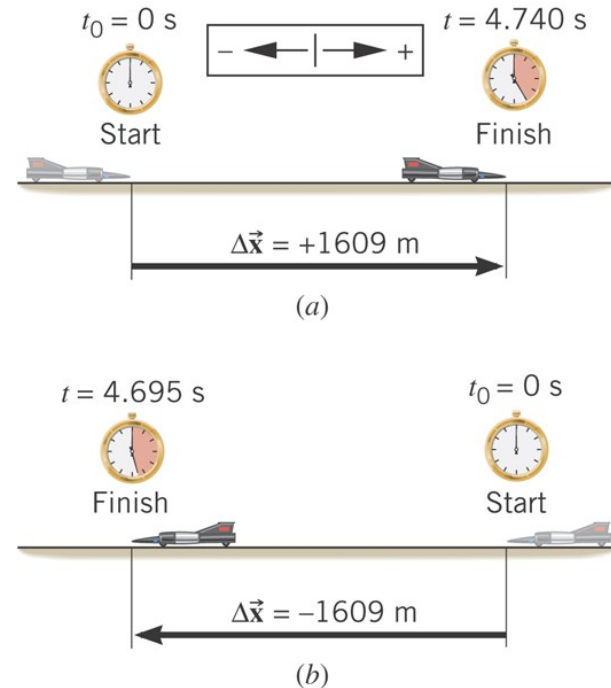
any quantity without a direction)



2.2 Speed and Velocity

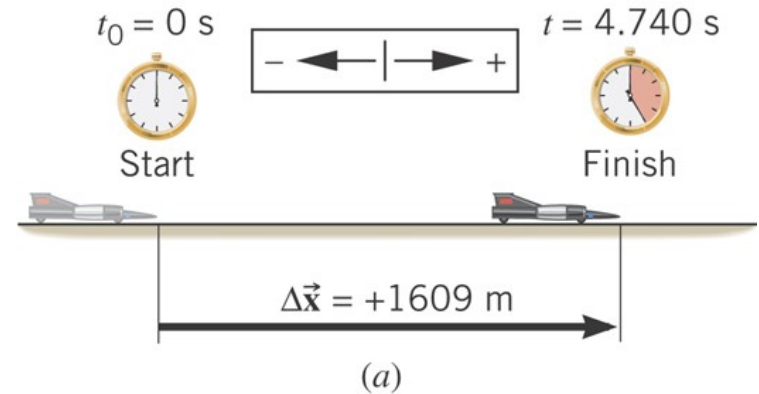
Example 2 The World's Fastest Jet-Engine Car

Andy Green in the car *ThrustSSC* set a world record of 341.1 m/s in 1997. To establish such a record, the driver makes two runs through the course, one in each direction, to nullify wind effects. From the data, determine the average velocity for each run.

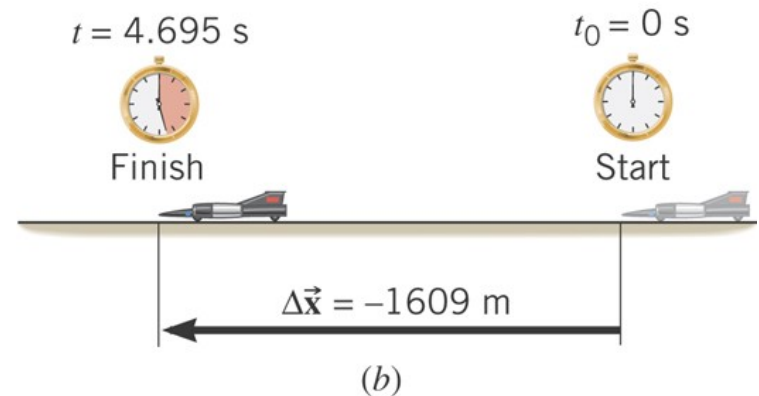


2.2 Speed and Velocity

$$\vec{v} = \frac{\Delta \vec{x}}{\Delta t} = \frac{+1609 \text{ m}}{4.740 \text{ s}} = +339.5 \text{ m/s}$$



$$\vec{v} = \frac{\Delta \vec{x}}{\Delta t} = \frac{-1609 \text{ m}}{4.695 \text{ s}} = -342.7 \text{ m/s}$$



Displacement = how far you are from your starting point and @ left or @ right.

Distance = total distance covered

1) A race car covers 1700m @ right in 5s. What is the average velocity?

The same race drive in the opposite direction with the wind opposing

Its motion. It covers 1700m @ left in 5.4 s. What is the average velocity ?

2) In NYC you walk 10 blocks @ south (or -10) in 30minutes. What is your average velocity in blocks/hour **In class**

3) you drive 100 miles @ east (so +10) in 2 hours.

What is your average velocity in mph and m/s

4) If you walk 5 blocks @ north and you realize this is the wrong way. **In class**

You then walk 10blocks @ south. The all trip lasted 30 minutes.

What is the average velocity (from start to end) blocks/hour

What is the average speed. (not the same)

5) same question for 10 blocks @ west (-10) and 20 blocks @ east.

The trip lasts 45 minutes

6) same question for 100 miles @ North and 50 miles @ south in 2 hours.

In mph in m/s

7) An insect flies 5 inches @ up 3 inches down then 15 inches @ up in 2 s.

What is its average velocity ? In in/s Average speed?

Δx = displacement

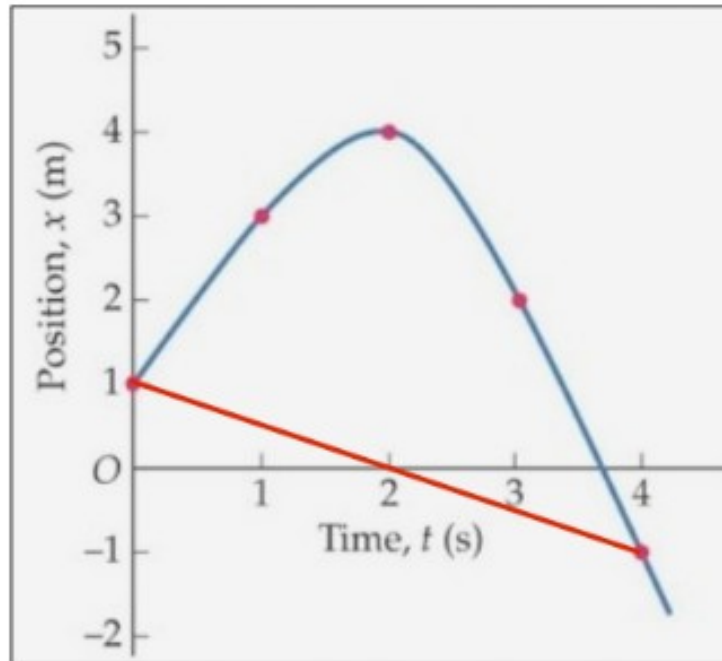
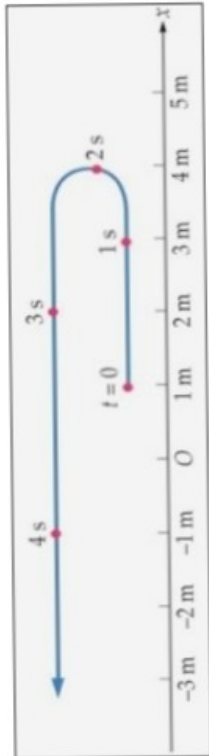
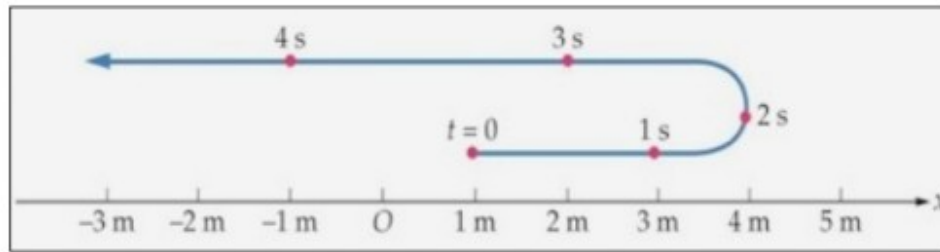
V = velocity

$V = \Delta x / t$

d = distance

S = speed

$S = d / t$

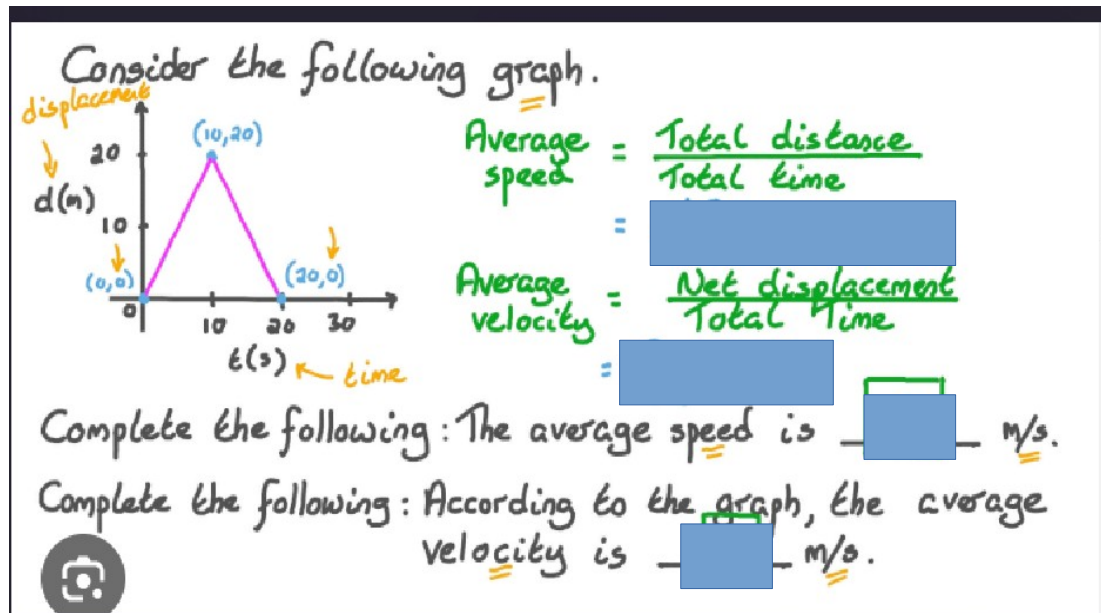


we transition from a number line graph
(position graph in space)
to a position vs. time graph $x(t)$
The graph is continuous.
Find average speed and
average velocity between
0 and 4s

So the average velocity is the slope of the line connecting the initial position
To the final position. (rise / run). Unit is m/s. Slope is negative.
Instantaneous velocity is the tangent at a given point

Average speed is the total distance / time elapsed.

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$



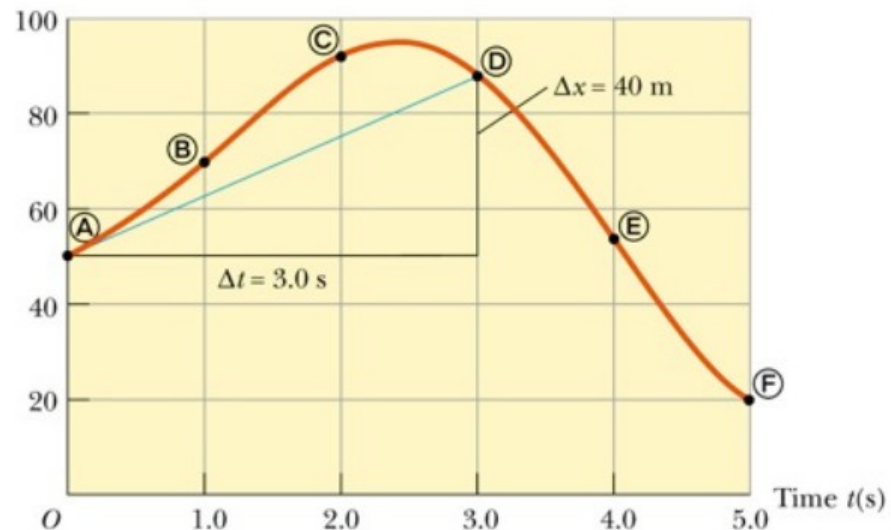
Make a number line diagram.

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$

Whats the average velocity

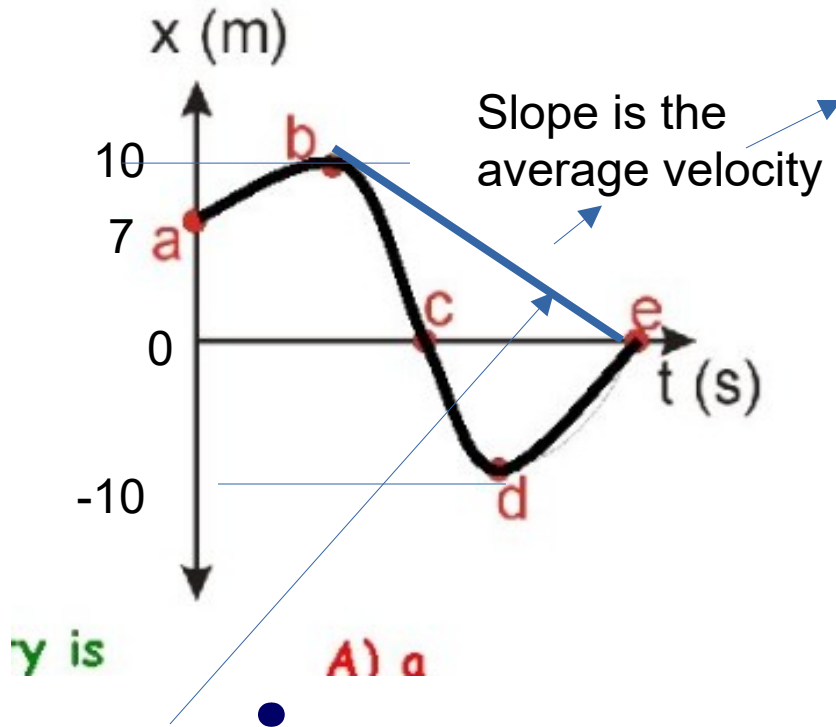
Position x(m)

© 2002 Brooks Cole Publishing - a division of Thomson Learning



Average Velocity

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$



Between b and e (b is at 2s and e is at 4s)

The displacement is negative

So the average velocity is negative.

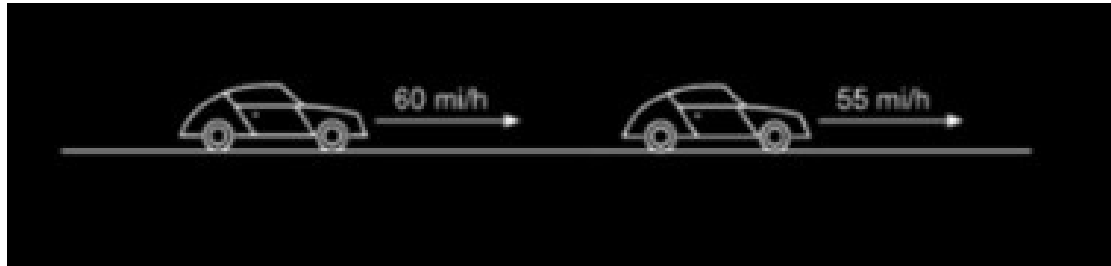
The slope is negative

This is not the instantaneous speed (dx/dt or tangent at a given time)

Between c and e the average velocity is 0

Between d and e the average velocity is positive

Relative velocity

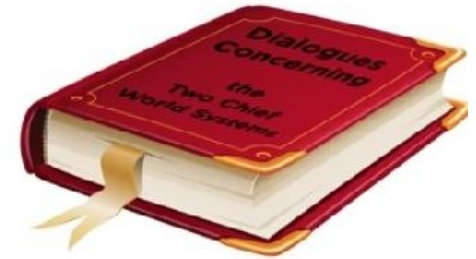
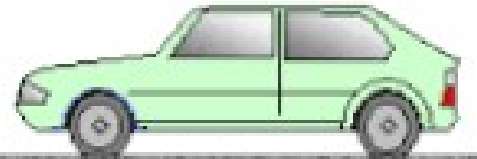


1

25 ms⁻¹

$$V_{\text{car}/u} = V_{\text{car}/\text{gnd}} - V_{u/\text{gnd}}$$

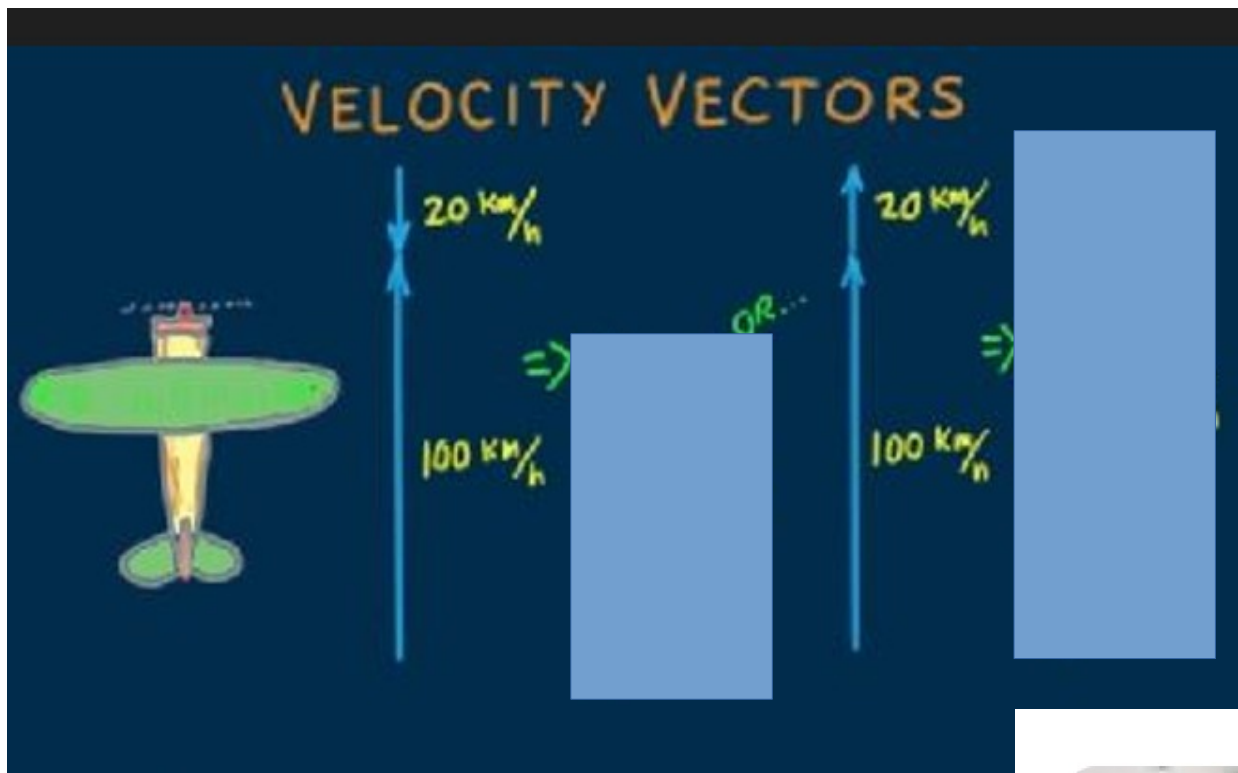
35 ms⁻¹



published in 1632

If the car in front of you is moving at the same velocity as you/
 Relative to you, its speed is / If the car in front of you slows down
 Relative to you, the car is moving toward? Away? Same distance?
 Example: sky diver opening a parachute filmed by a sky diver without
 a chute.

Dialogue Concerning the Two Chief World Systems was a revolutionary book about the science of mechanics. It's written as a conversation and, in it, Galileo's three characters spend much of the second day discussing relative motion, including whether or not the Earth is stationary. "It is impossible by mechanical means to say whether we are moving or staying at rest"



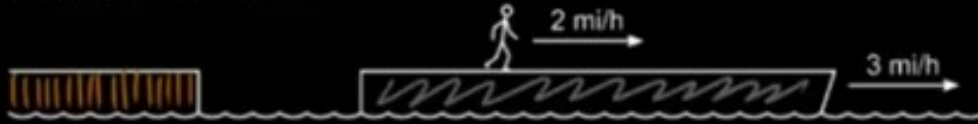
$$V_{\text{plane/gnd}} = V_{\text{plane/wind}} + V_{\text{wind/gnd}}$$



➡ Swimming by Skills NT
Counter-current Pools: Alternative to...

Relative velocity

A barge leaves the dock, heading away at 3 mi/h. A man on the barge, facing away from the dock, walks at 2 mi/h relative to the barge. What is the man's speed relative to the dock?



$$V_{\text{man/dock}} = V_{\text{man/boat}} + V_{\text{boat/dock}}$$

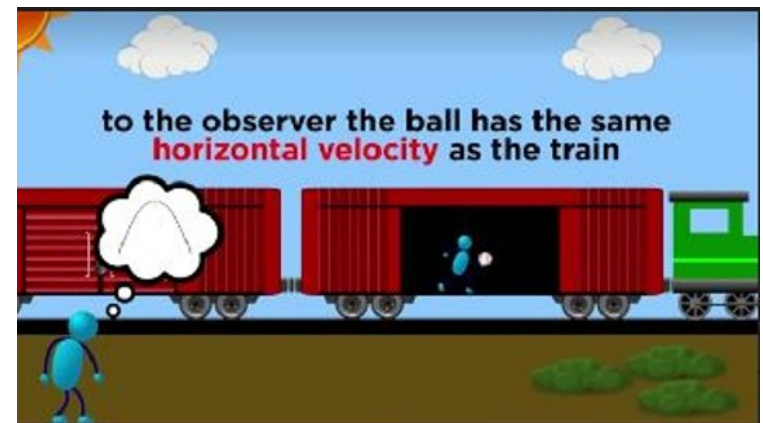
A barge leaves the dock, heading away at 3 mi/h. A man on the barge, facing toward from the dock, walks at 2 mi/h relative to the barge. What is the man's speed relative to the dock?



$$V_{\text{man/dock}} = V_{\text{man/boat}} + V_{\text{boat/dock}}$$

We already discussed frame of reference.

The retrograde motion of Mars is another Example.(picture)



Discuss relative speed if both train and person has the same speed but opposite See video

2.2 *Speed and Velocity*

The ***instantaneous velocity*** indicates how fast the car moves and the direction of motion at each instant of time.

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t}$$

Example 2.1 textbook (university Physics)

- This plot shows the average velocity being measured over shorter and shorter intervals.

Instantaneous velocity is the slope of the tangent to the curve at the time of interest

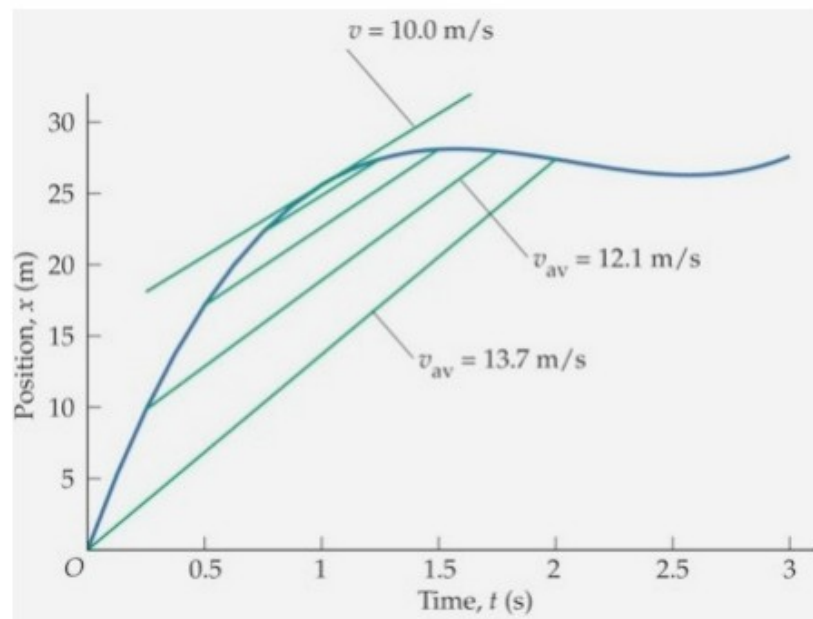
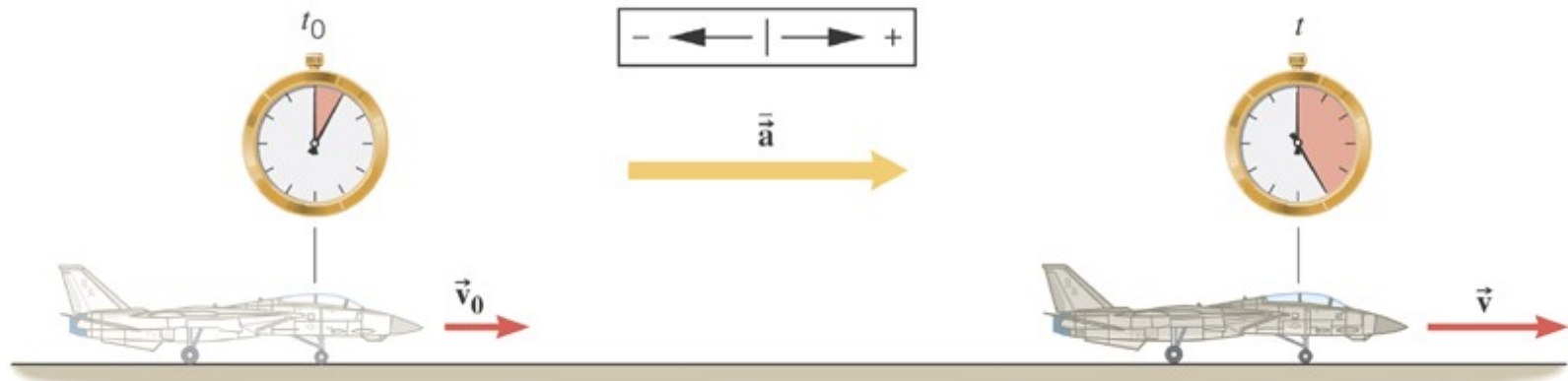


TABLE 2-2 Calculating the Instantaneous Velocity at $t = 1$ s.

t_i (s)	t_f (s)	Δt (s)	x_i (m)	x_f (m)	Δx (m)	$v_{av} = \Delta x / \Delta t$ (m/s)
0	2.00	2.00	0	27.4	27.4	13.7
0.250	1.75	1.50	9.85	28.0	18.2	12.1
0.500	1.50	1.00	17.2	28.1	10.9	10.9
0.750	1.25	0.50	22.3	27.4	5.10	10.2
0.900	1.10	0.20	24.5	26.5	2.00	10.0
0.950	1.05	0.10	25.1	26.1	1.00	10.0

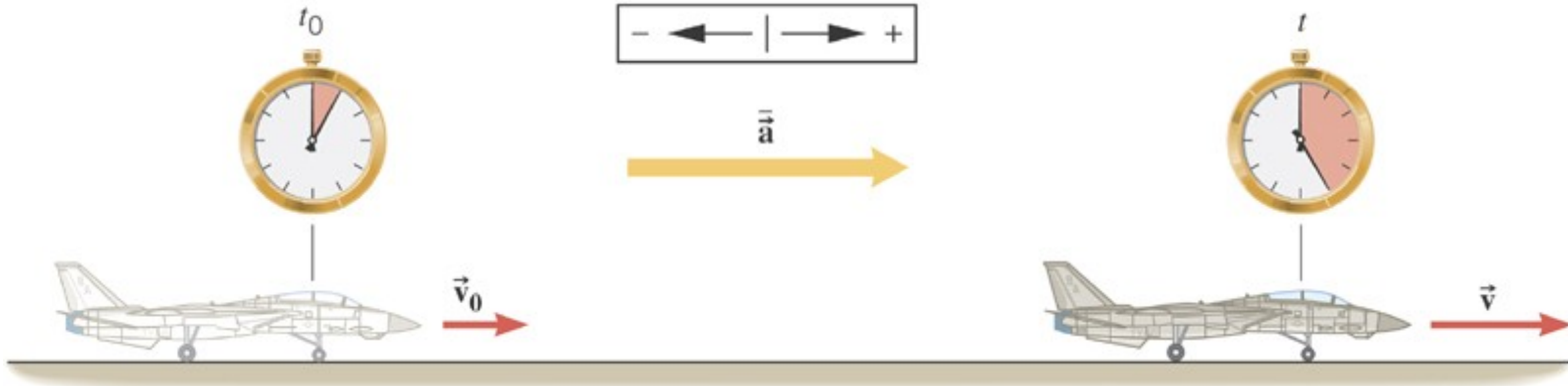
2.3 Acceleration

The notion of *acceleration* emerges when a change in velocity is combined with the time during which the change occurs.



Example: $g = 22 \text{ mph/s}$ or 32ft/s/s

2.3 Acceleration



DEFINITION OF AVERAGE ACCELERATION

$$\vec{a} = \frac{\vec{v} - \vec{v}_o}{t - t_o} = \frac{\Delta \vec{v}}{\Delta t}$$



Is the speed constant or what?

You can find the average the
Average acceleration

How to find the average velocity?

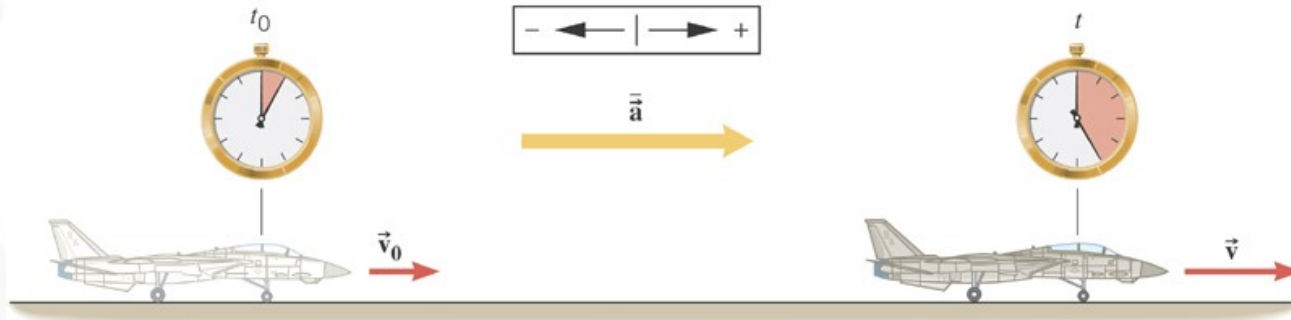
Calculus minute.
How to find

mean, $\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$

ACCELERATION = $\frac{\text{CHANGE IN SPEED}}{\text{TIME}}$

$\frac{31.5 \text{ ft/s}}{1.2 \text{ sec}}$

ACCELERATION = 26.25 ft/s^2



Example Acceleration and Increasing Velocity

Determine the average acceleration of the plane in km/h/s

$V_{\text{initial}} = 0$

$V_{\text{final}} = 250 \text{ km/h}$

time elapsed = 30s

A) Acceleration in km/h/s (see next slide before going to B)

B) How fast it is going after 5 seconds (find speed in km/h)

C) for the same acceleration what is the speed after 5s if the initial velocity is 50km/h ?

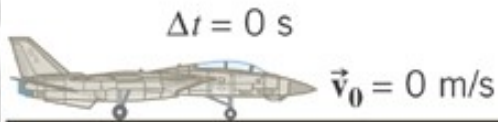
31

D) convert the speed found in B to m/s (1km=1000m and 1 hour=3600s)

1m/s = 3.6 km/h

$$\vec{a} = + 8 \frac{\text{km/h}}{\text{s}}$$

Can you find the rate of Increase?



Acceleration is a vector:
8 is the magnitude ; km/h/s is the unit ; and + is the direction

If there is no acceleration ($a=0$)
 → the velocity is constant

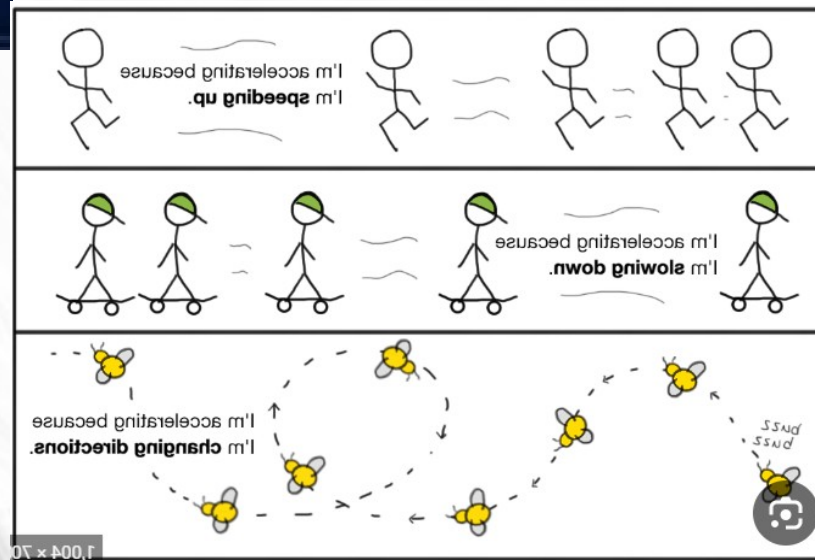
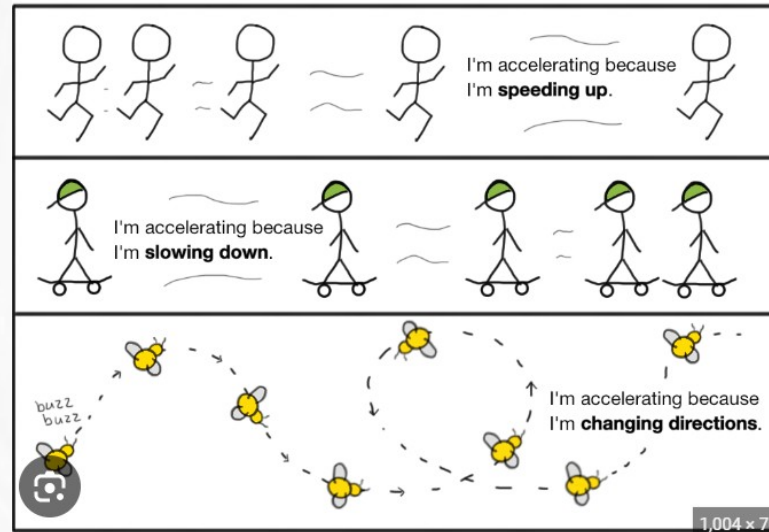
Velocity and Acceleration

Do not need to be in same direction

Acceleration in same direction
 as velocity increases speed

Acceleration opposite velocity
 decreases speed

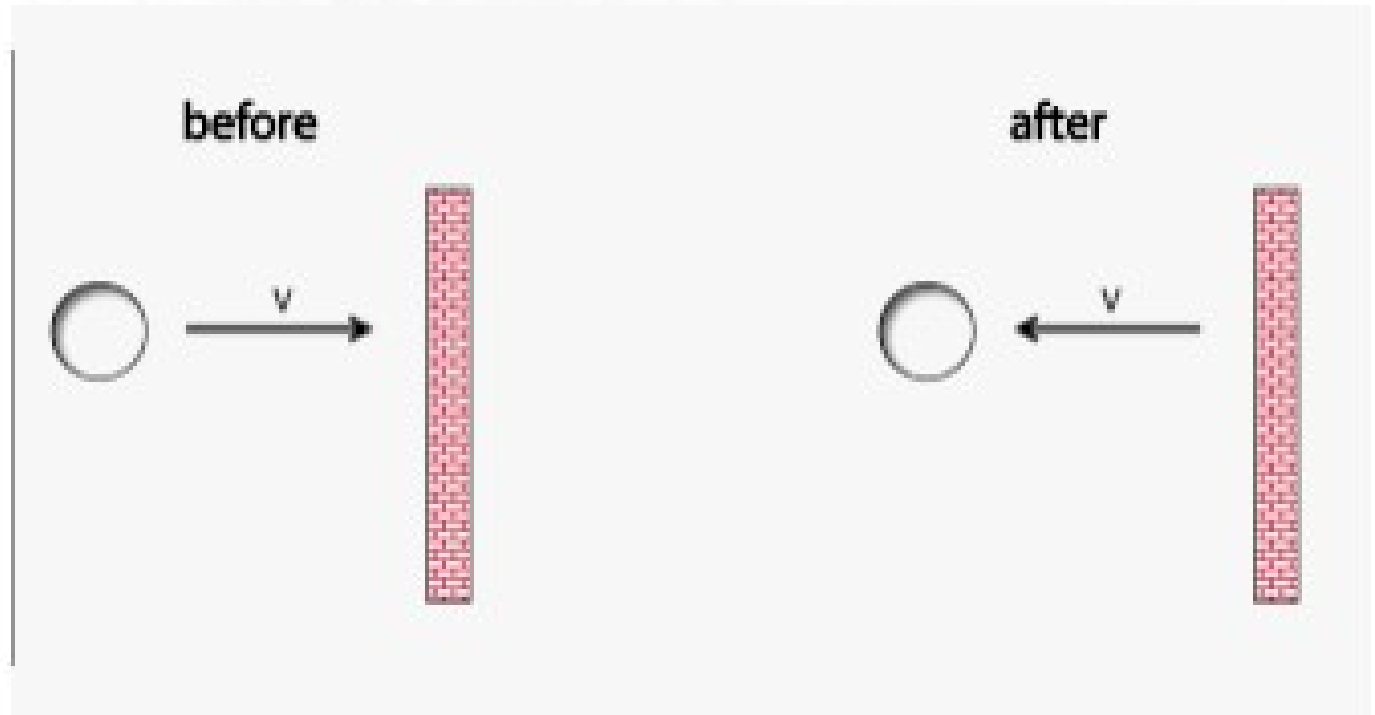
Acceleration at right angles to
 velocity changes only direction



The acceleration is in the same
 direction as the force!

Which way velocity is?
 Acceleration?
 Force?
 In all those cases.

Bouncing ball what is the acceleration ($v=5\text{m/s}$ in magnitude) in 0.01 s . use definition
Which way does the force point?



http://www.walter-fendt.de/html5/phen/acceleration_en.htm

$a=-1$ show anatomy

Acceleration and Decreasing Velocity

Initial velocity = 28 m/s

Final velocity = 13 m/s

Between $t=9\text{ s}$ and $t=12\text{ s}$

Find the acceleration

1. 5 m/s/s

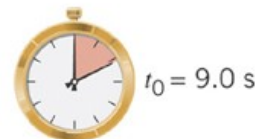
2. 1.7 m/s/s

3. -5 m/s/s

The acceleration has the same direction
than the force (push/pull)

Here the pull is @ left so acceleration is @ ____
but the velocity is @ _____

Velocity is a vector too. It has a _____
And a _____.

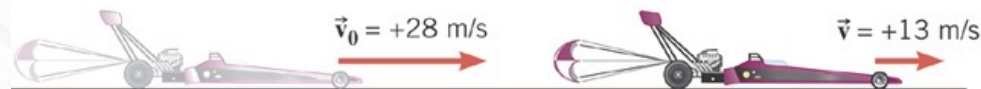
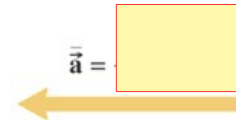


$t_0 = 9.0\text{ s}$

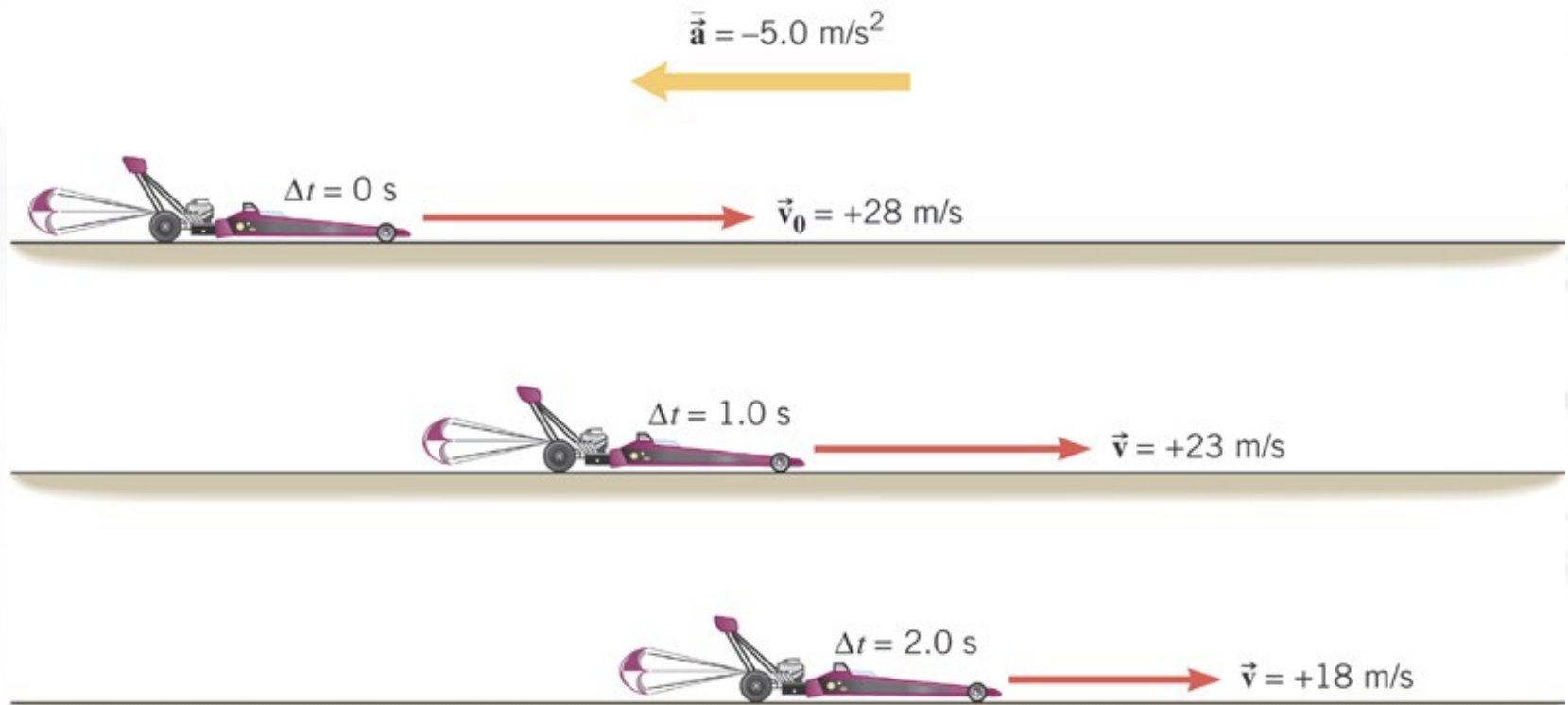


$t = 12\text{ s}$

$\vec{a} =$



(b)



If velocity and acceleration have the same direction => speeding up

If velocity and acceleration have opposite direction => slowing down

http://www.walter-fendt.de/html5/phen/acceleration_en.htm

Introduce equations of kinematics.

Try $a = 0$ $V_0 = 5$ $X_0 = 0$

$A = 0$ $V_0 = 5$ $X_0 = 10$

Use $V_0 = 5$ and $a = 1$

Then $V_0 = 10$ and $a = -1$

Try also $V_0 = 0$ $a = -1$ $X_0 = 50$

2.4 *Equations of Kinematics for Constant Acceleration*

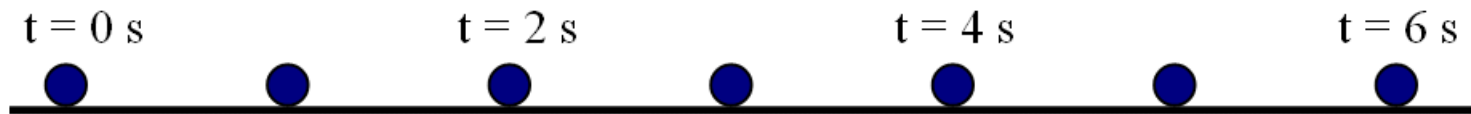
Five kinematic variables:

1. displacement, x
2. acceleration (constant), a
3. final velocity (at time t), v
4. initial velocity, v_o
5. elapsed time, t

Starting from rest, a particle confined to move along a straight line is accelerated at a rate of 2 m/s^2 .

- a) The particle travels 2 m during each second.
- b) The particle travels 2 m only during the first second.
- c) The speed of the particle increases by 2 m/s during each second.
- d) The acceleration of the particle increases by 2 m/s^2 during each second.

The drawing shows the position of a rolling ball at one second intervals. Which one of the following phrases best describes the motion of this ball?



- a) constant position
- b) constant velocity
- c) increasing velocity
- d) constant acceleration
- e) decreasing velocity

In which one of the following situations does the car have an acceleration that is directed due north?

- a) A car travels northward with a constant speed of 24 m/s.
- b) A car is traveling southward as its speed increases from 24 m/s to 33 m/s.
- c) A car is traveling southward as its speed decreases from 24 m/s to 18 m/s.
- d) A car is traveling northward as its speed decreases from 24 m/s to 18 m/s.

A postal truck driver driving due east gently steps on her brake as she approaches an intersection to reduce the speed of the truck. What is the direction of the truck's acceleration, if any?

- a) There is no acceleration in this situation.
- b) due north
- c) due east
- d) due south
- e) due west

A sports car starts from rest. After 10.0 s, the speed of the car is 25.0 m/s. What is the magnitude of the car's acceleration?

- a) 2.50 m/s^2
- b) 5.00 m/s^2
- c) 10.0 m/s^2
- d) 25.0 m/s^2
- e) 250 m/s^2

Which one of the following is not a vector quantity?

- a) acceleration
- b) displacement
- c) instantaneous velocity
- d) average velocity
- e) average speed

Which one of the following equations is the correct expression for average acceleration?

a) $a = \frac{v}{t}$

b) $a = \frac{dv}{dt}$

c) $a = \frac{\Delta x}{\Delta t}$

d) $a = \frac{\Delta v}{\Delta t}$

e) $a = \frac{1}{2} vt^2$

e)

1) What is the acceleration of a rocket ship in outer space that takes 5.0s to increase its speed from 1240m/s to 1300m/s ?
(textbook)

2) In outer space a rocket ship is traveling at the enormous speed of 2800m/s. What is its acceleration if it increases its speed uniformly and is going 2840m/s after 25 s ?
Convert to mph (do it in class)

3) A car travels with an average speed of 25m/s (do it in class)
A) What is the speed in km/s (1km =1000m)
B) What is the speed in km/h (1m/s = 3.6 km/h)

4) A car travels with an average speed of 58mph. What is the speed in km/h (1.6km = 1 mile)

5) Starting from rest and moving in a straight line, a runner achieves a velocity of 7m/s in a time of 2s. What is the average acceleration of the runner ?

6) starting from rest, a car accelerates at a rate of 4.2m/s/s for a time of 5 seconds.
What is its velocity at the end of this time ?

7) The velocity of a car decreases from 30m/s to 18m/s in a time of 4seconds. What is the average acceleration if the car
In this process ?

Equations of kinematics: the big 5

Notations:

V (V_{final}) final velocity

V_i (V_{initial}) initial velocity $\rightarrow$ velocity at $t=0$

X_i (X_{initial}) initial position $\rightarrow$ position at $t=0$

X (X_{final}) final position

Δx is the displacement ($x-x_i$) how far you are from "home"

a acceleration

t is the time elapsed

V_i is the initial velocity and is
Sometimes noted V_0

X_i is the initial position (home)
And is sometimes noted x_0

EQUATIONS OF KINEMATICS

for motion with constant acceleration

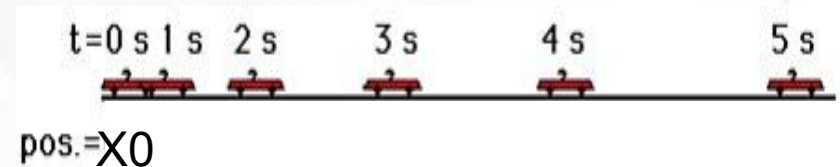
1) $x = x_i + V_i t + 0.5 a t^2$ or $\Delta x = V_i t + 0.5 a t^2$ if $x_i=0$ then $x = V_i t + 0.5 a t^2$

2) $V = V_i + at$

3) $a = (V - V_i)/t$

4) $V^2 = V_i^2 + 2a(x - x_i)$ or $V^2 = V_i^2 + 2a(\Delta x)$

5) Average velocity $= \Delta x/t$ or $(V + V_i)/2$



EQUATIONS OF KINEMATICS

for motion with constant velocity and acceleration $= 0$

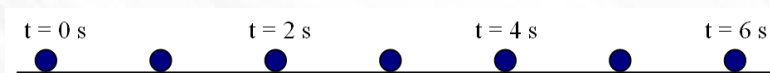
1) $x = x_i + V_i t$ or $\Delta x = V_i t$ if $x_i=0$ then $x = V_i t$

2) $V_i = V_f$ stays the same

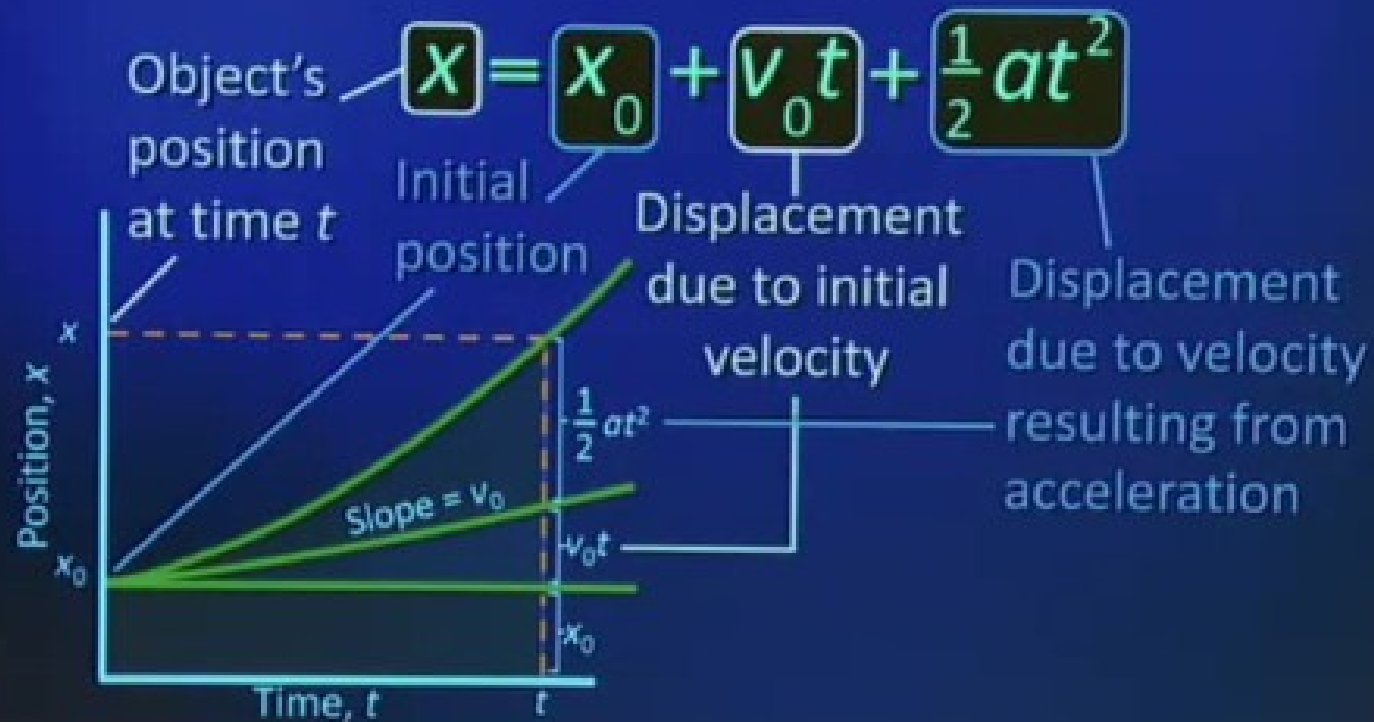
3) $a = 0$

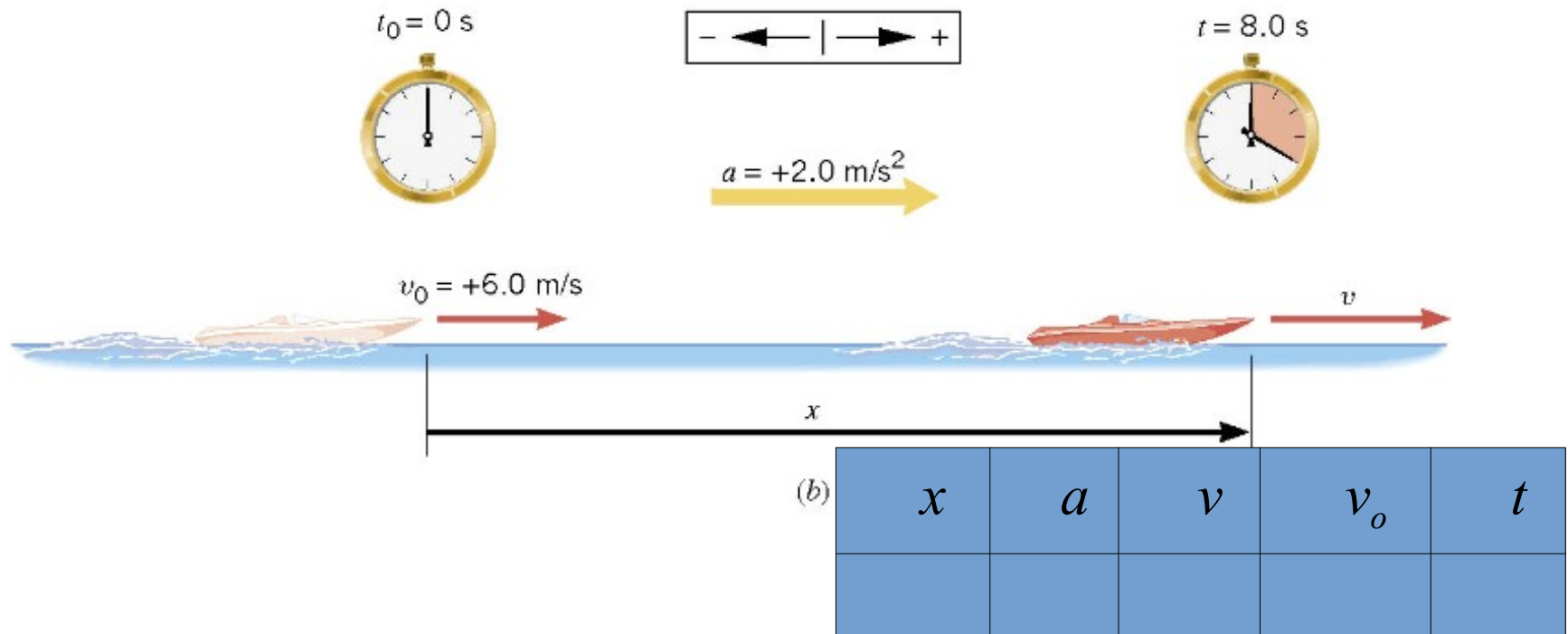
4) NA

5) Average velocity $= \Delta x/t$



Anatomy of an Equation

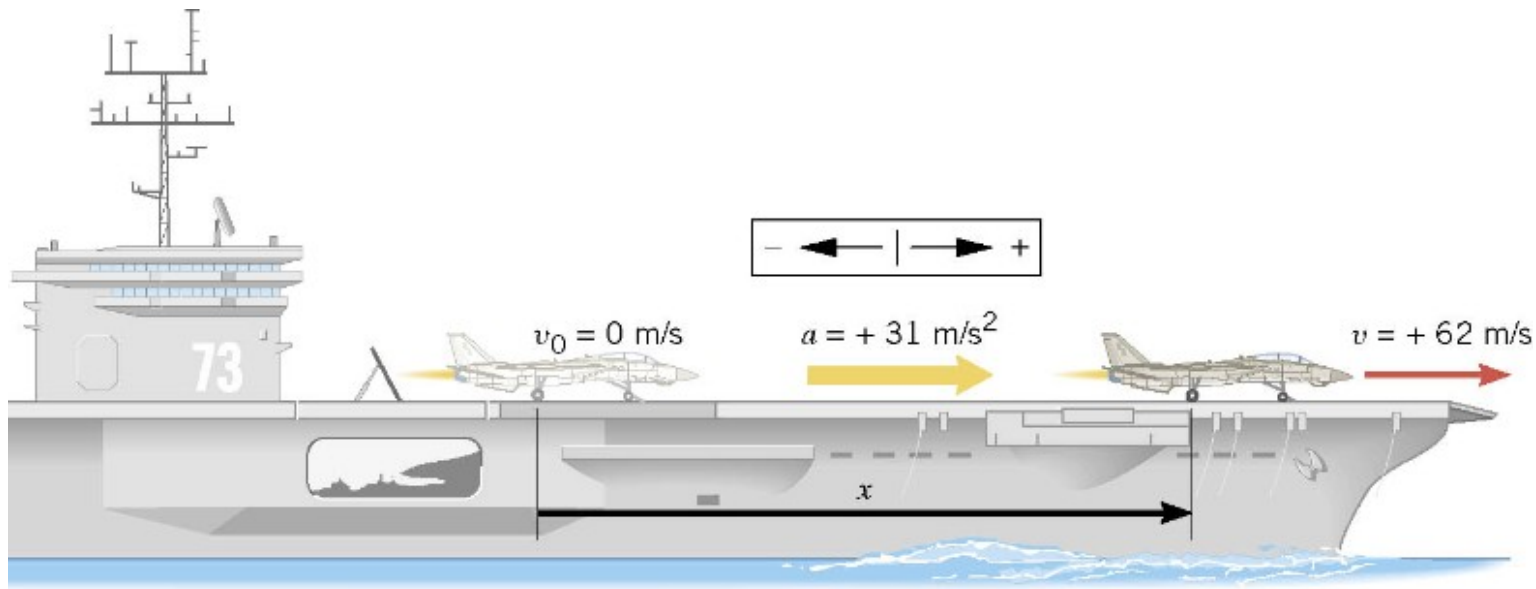




FIND THE displacement of the boat :

1. 64m
2. 72m
3. 76m
4. 112m

[1]	$x = x_0 + v_0 t + \frac{1}{2} a t^2$
[2]	$v = v_0 + a t$
[3]	$a = \text{constant}$
[4]	$v^2 = v_0^2 + 2a(\Delta x)$



FIND THE displacement ^(b) of the airplane :

1. 62m

2. 1m

3. 124m

x	a	v	v_o	t

Example 2.4

[1] $x = x_0 + v_0 t + \frac{1}{2} a t^2$

[2] $v = v_0 + a t$

[3] $a = \text{constant}$

[4] $v^2 = v_0^2 + 2a(\Delta x)$

2.5 *Applications of the Equations of Kinematics*

Reasoning Strategy

1. Make a drawing.
2. Decide which directions are to be called positive (+) and negative (-).
3. Write down the values that are given for any of the five kinematic variables.
4. Verify that the information contains values for at least three of the five kinematic variables. Select the appropriate equation.
5. When the motion is divided into segments, remember that the final velocity of one segment is the initial velocity for the next.
6. Keep in mind that there may be two possible answers to a kinematics problem.

Example 8 An Accelerating Spacecraft

A spacecraft is traveling with a velocity of +3250 m/s. Suddenly the retrorockets are fired, and the spacecraft begins to slow down with an acceleration whose magnitude is 10.0 m/s². What is the velocity of the spacecraft when the displacement of the craft is +215 km, relative to the point where the retrorockets began firing

x	a	v	v_o	t

2 possible answers.

- 1. +/- 6262500 m/s**
- 2. +/- 2502.5 m/s**
- 3. +/- 3855 m/s**

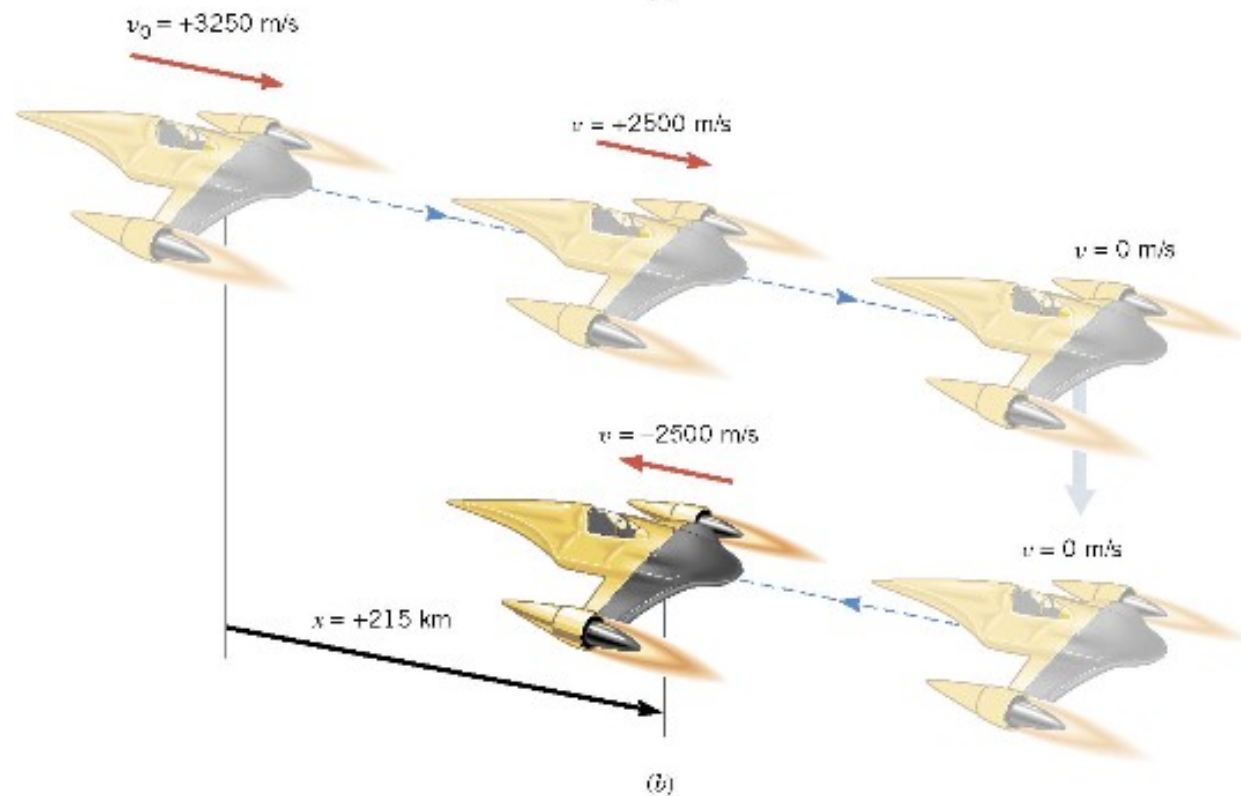
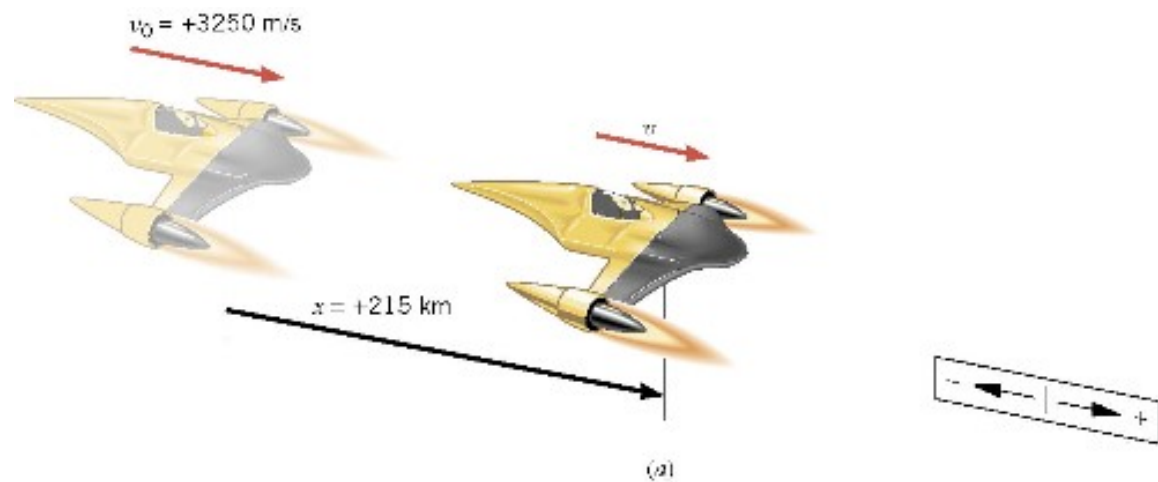
$$[1] \quad x = x_0 + v_0 t + \frac{1}{2} a t^2$$

$$[2] \quad v = v_0 + a t$$

$$[3] \quad a = \text{constant}$$

$$[4] \quad v^2 = v_0^2 + 2a(\Delta x)$$

2.5 Applications of the Equations of Kinematics



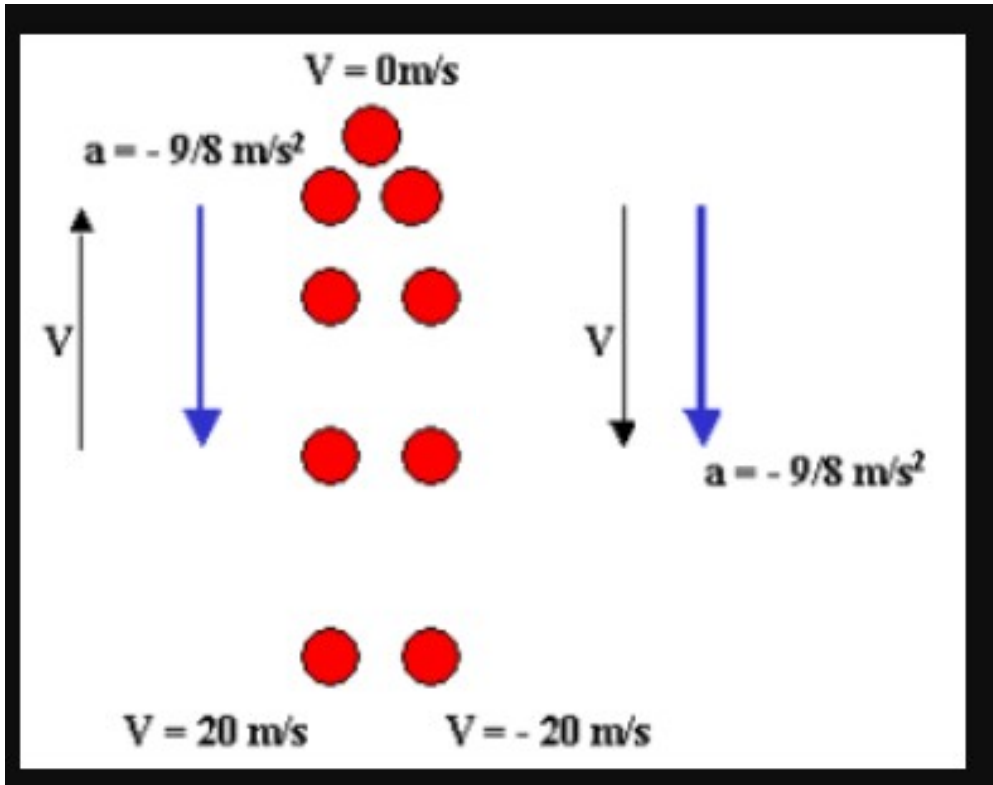
2.5 Applications of the Equations of Kinematics

x	a	v	v_o	t
+215000 m	-10.0 m/s ²	?	+3250 m/s	

$$v^2 = v_o^2 + 2ax \longrightarrow v = \sqrt{v_o^2 + 2ax}$$

$$\begin{aligned} v &= \pm \sqrt{(3250 \text{ m/s})^2 + 2(10.0 \text{ m/s}^2)(215000 \text{ m})} \\ &= \pm 2500 \text{ m/s} \end{aligned}$$

Similar situation:



Throwing a ball straight up.

The weight (force) is always pulling @ down. So acceleration is always @ down (negative)

On the way up the force causes The ball to slow down.

On the way down the force causes The ball to speed up.

Find the kinematics equation

**Let's use the 2 equations : $\Delta x = v_i t + 0.5 (a) t^2$ and $V = v_i + at$
or $x = x_i + v_i t + 0.5 (a) t^2$**

1) A car accelerating with an initial velocity of 12m/s accelerates at a constant rate of 2.5m/s/s for a time of 2s.

A) What is its velocity at the end of this time ?

B) What displacement does the car cover during this process (the initial position $X_o = 0$ find X)

2) A car moving with an initial velocity of 30m/s slows down at a constant rate of -3m/s/s

A) What is the velocity after 3 seconds of deceleration

B) What displacement does the car cover in this time: (Find X with $X_o = 0$)

3) A runner moving with an initial velocity of 9m/s slows down at a constant rate of -1.5m/s/s over a period of 2 seconds.

A) What is her velocity at the end of this time

B) What displacement did she cover during this process(X with $X_o = 0$)

4) if $V_o = 3\text{m/s}$ and $V_{\text{final}} = 9\text{m/s}$ What is the average velocity if the acceleration is constant.

6) If the initial velocity of a car is 30m/s and its acceleration is 10m/s/s . Initial position is 0.

A) what its equation of motion $x=f(t)$.

B) what is the equation of $V=f(t)$ (velocity as a function of time)

7) if the equation of motion of an object is $x = 5 + 10t + 2t^2$ What are:
Initial position ? initial velocity ? Acceleration ?

8) same question for $x = 10t + t^2$

9) $x = 10t^2$

10) if $x = 5 + 10t + 10t^2$

What is the velocity at $t=5s$ (after 5s)

11) if $V = 10 + 2t$

What is the acceleration ?

What is the final velocity when $t=100s$ (so after 100s)?

What is the average speed during that time ?

1) How far does a car travel if it starts at rest and accelerated at 3m/s/s for **6.5s**

2) How long does it take a car, starting from rest, to travel 240m if its acceleration is 1.90m/s/s ? (in class)

3) How far a car travel if it starts at rest and accelerated at 3m/s/s until it reaches a speed of 22m/s ? (in class)

4) A car moving 22m/s has its brakes jammed on and leaves skid marks 45m long. What was its acceleration ?
(negative) (in class)

5) What is the acceleration of a rocket-driven sled that travels 360m in 5s , starting from rest and accelerating uniformly ? (HW)

HW

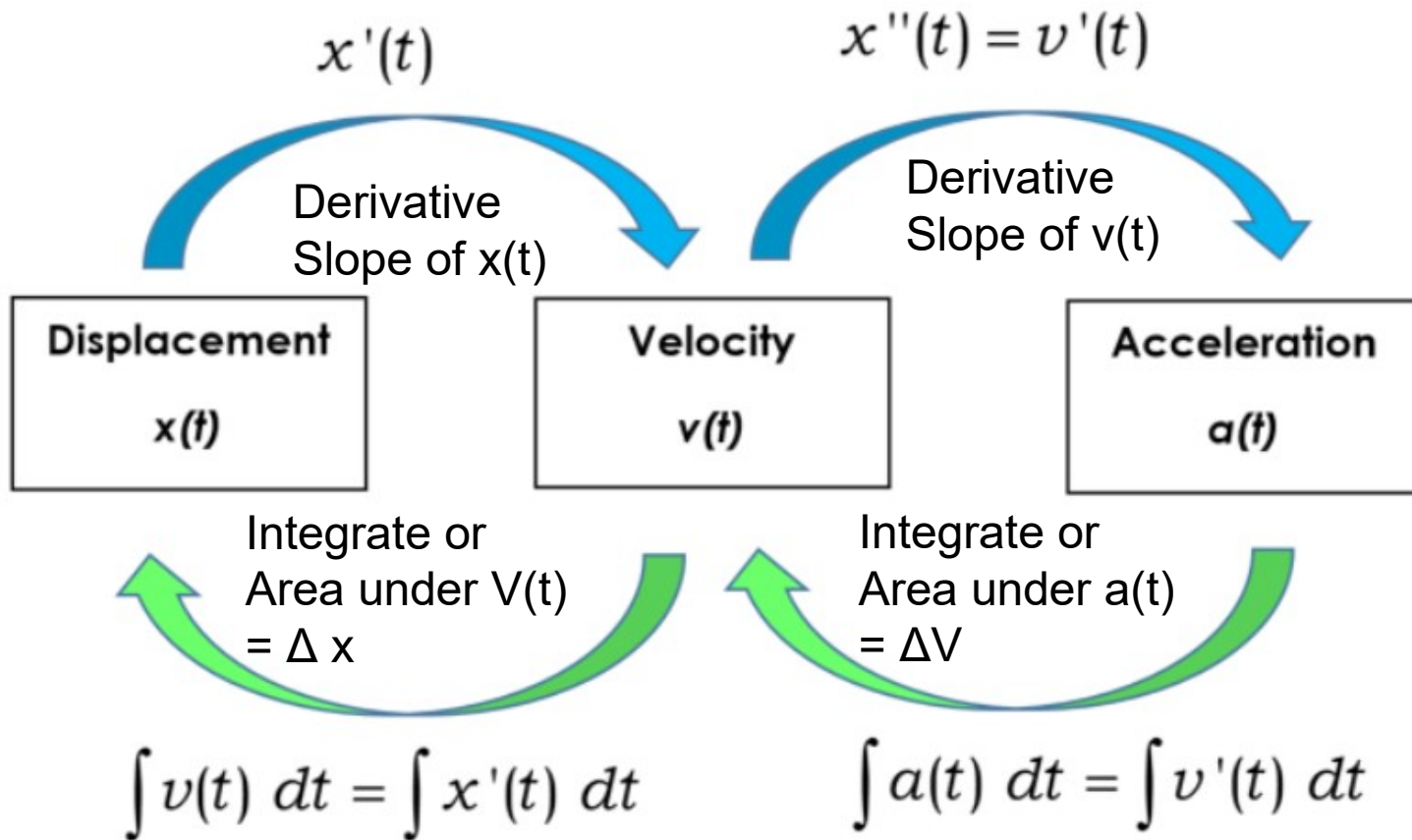
6) How fast is a car going if it starts at rest and accelerates uniformly at 2.8m/s/s while traveling 220m ? (HW)

7) A car traveling in a straight line with an initial velocity of 14m/s accelerates at 2m/s/s to a velocity of 24m/s .

What is the distance covered by the car in this process ?

What is the time elapsed.

Hint: to avoid solving quadratic equation, find t first. Using $a = (V-V_0)/t$



$$v = \frac{dx}{dt}$$

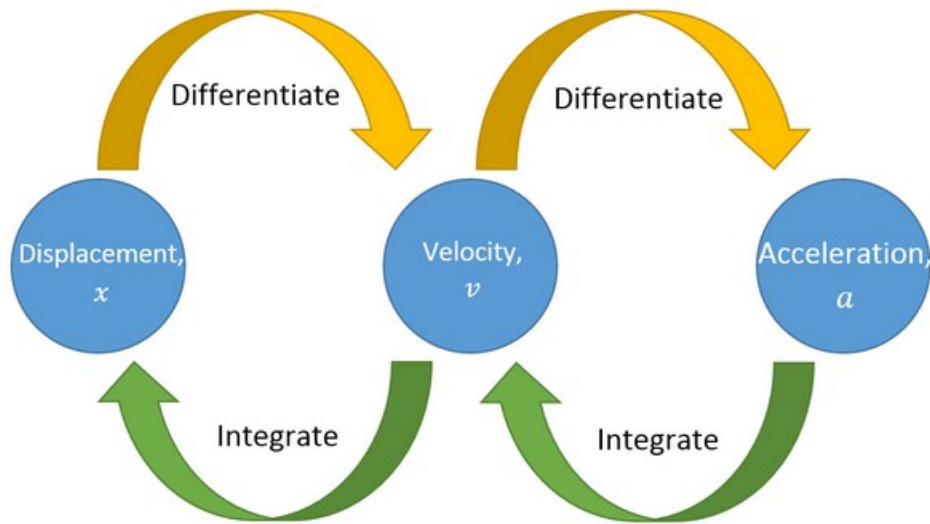
$$a = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2x}{dt^2}$$

$\mathbf{r}$ Position

$\frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$ Velocity (speed)

$\frac{d^2\mathbf{r}}{dt^2} = \ddot{\mathbf{r}}$ Acceleration

$\frac{d^3\mathbf{r}}{dt^3} = \dddot{\mathbf{r}}$ Jerk



$$v_x \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

A particle P is moving along the x -axis. The displacement x m from O can be written as a function of t

$$x = t^5 - 12t^2 + 9.$$

Find the formula for acceleration.

Solution
 $20t^3 - 24$

Worked Example: The Displacement at $t = 10$

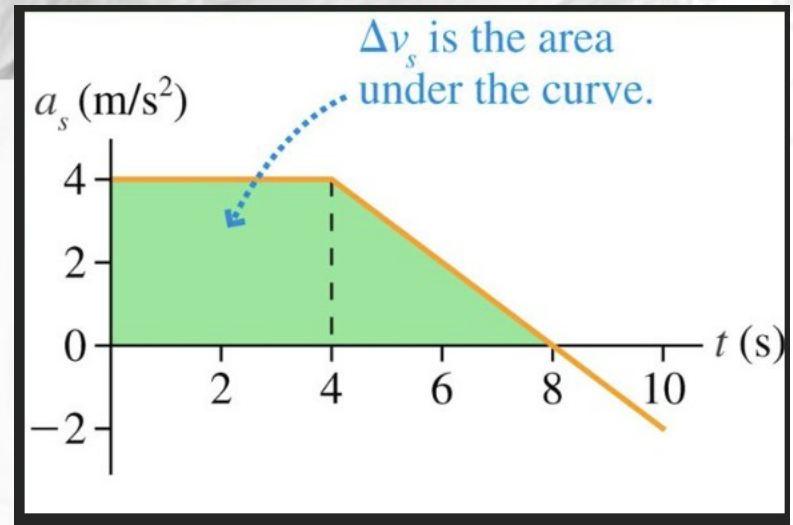
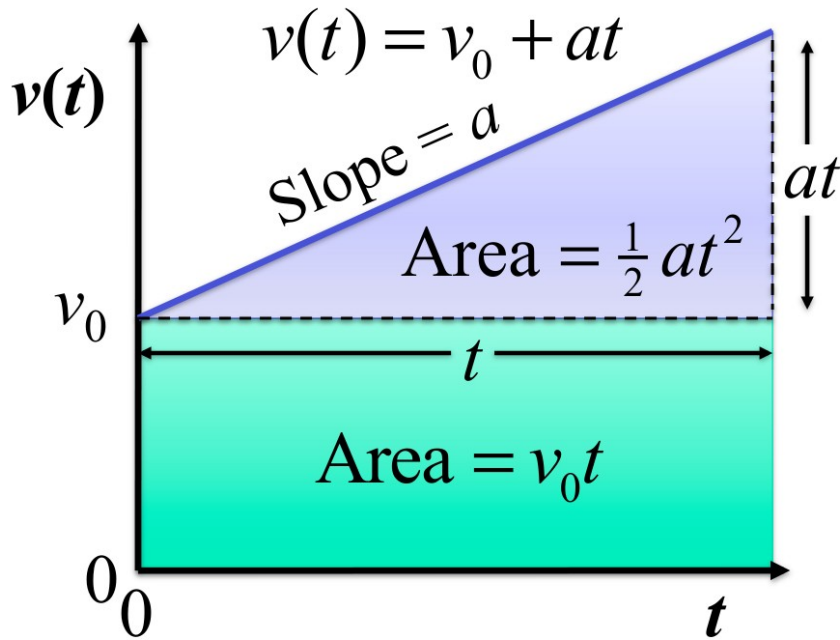
A particle P is moving along the x -axis. The acceleration $a \text{ ms}^{-2}$ from O can be written as a function of t

$$a = t + 19.$$

Suppose we know that when $t = 0$, we have $x = 0$ and when $t = 4$ we have $x = 10$. Find the displacement when $t = 10$.

Solution is $t^3/6 + 19/2 t^2 - 38t$ and $x(10) = 737 \text{ m}$

Constant acceleration: a special case



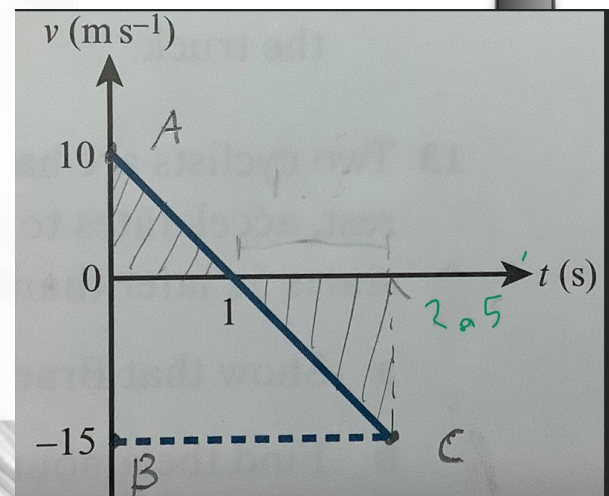
$$\Delta v = \int_{v_0}^v dv = \int_0^t a dt :$$

$$\Delta x = \int_{x_0}^x dx = \int_0^t v(t) dt = \int_0^t (v_0 + at) dt = \text{Area}$$



$$x - x_0 = v_0 t + \frac{1}{2} at^2$$

Example:
Displacement =
 $0.5(1)(10) - 0.5(1.5)(15) \text{ m}$



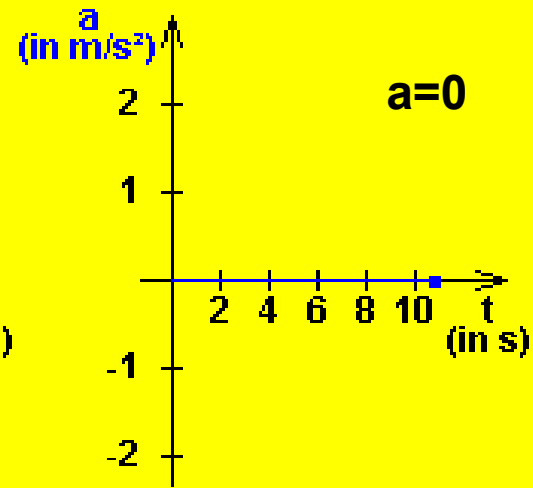
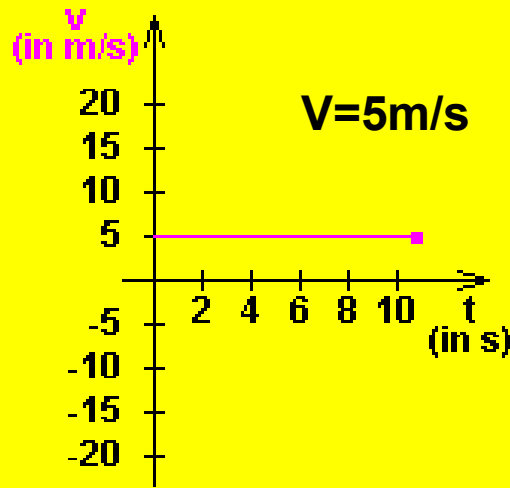
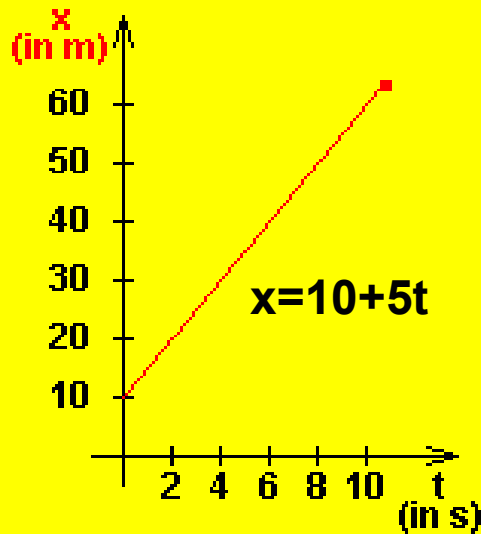
GRAPHICAL ANALYSIS OF MOTION

See applets/mechanics/acceleration_en

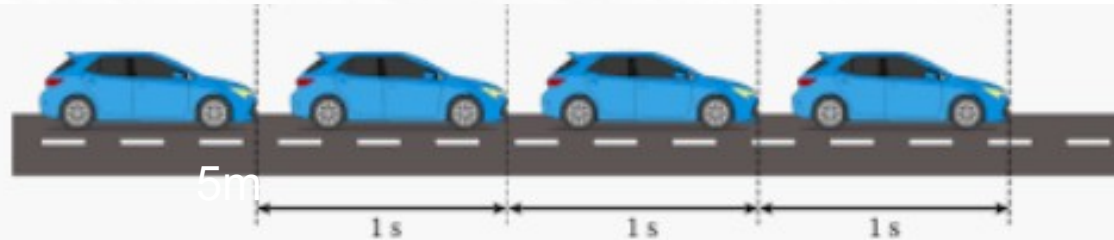
EQUATIONS OF KINEMATICS

Moving at right @ constant speed

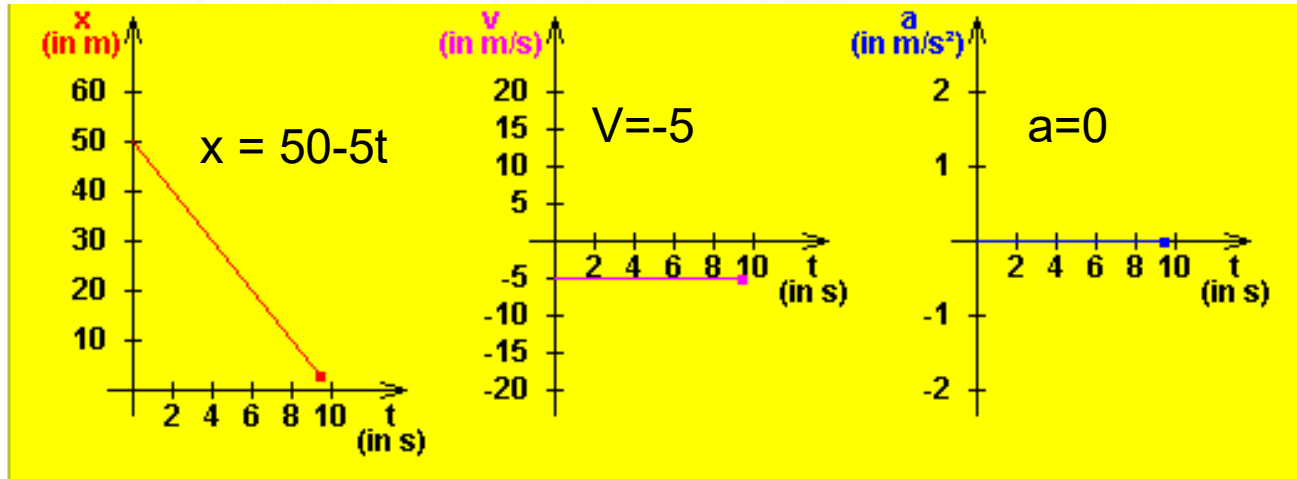
for motion with velocity and acceleration = 0



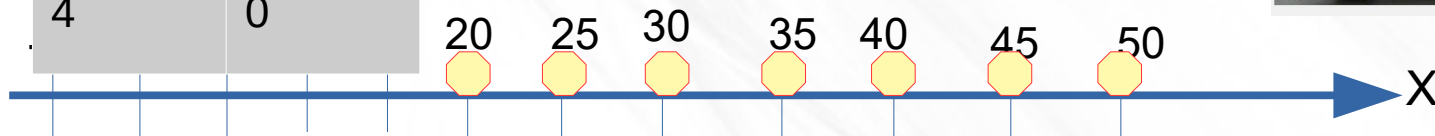
t(s)	x(m)
0	10
1	15
2	20
3	25
4	30



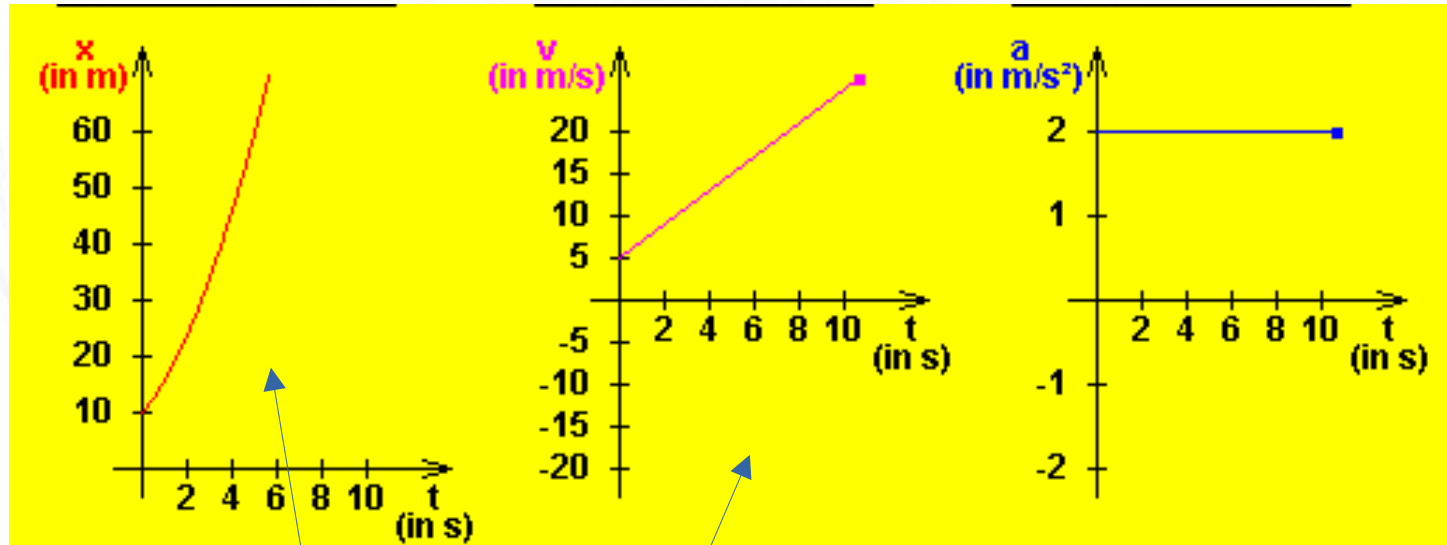
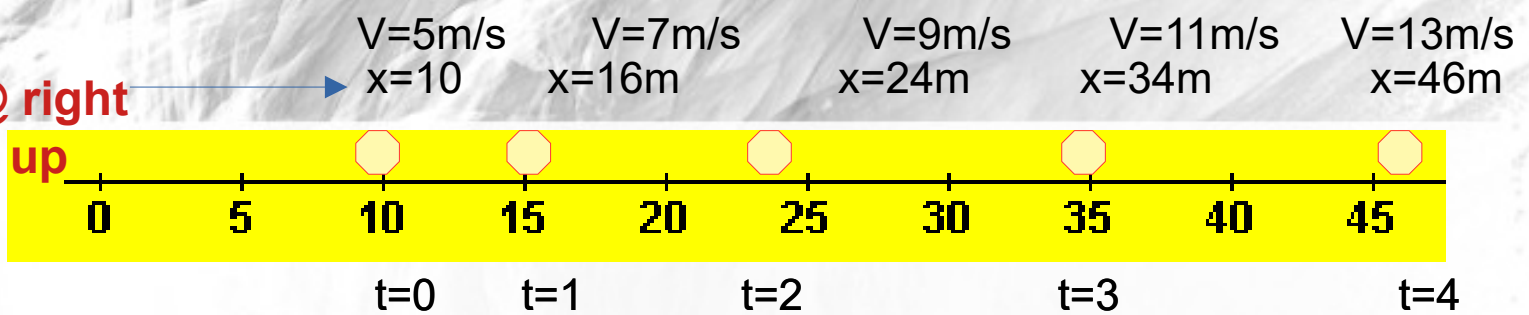
Moving @ left at constant speed



t(s)	x(m)
0	50
1	45
2	40
3	35
4	0



Moving @ right
Speeding up



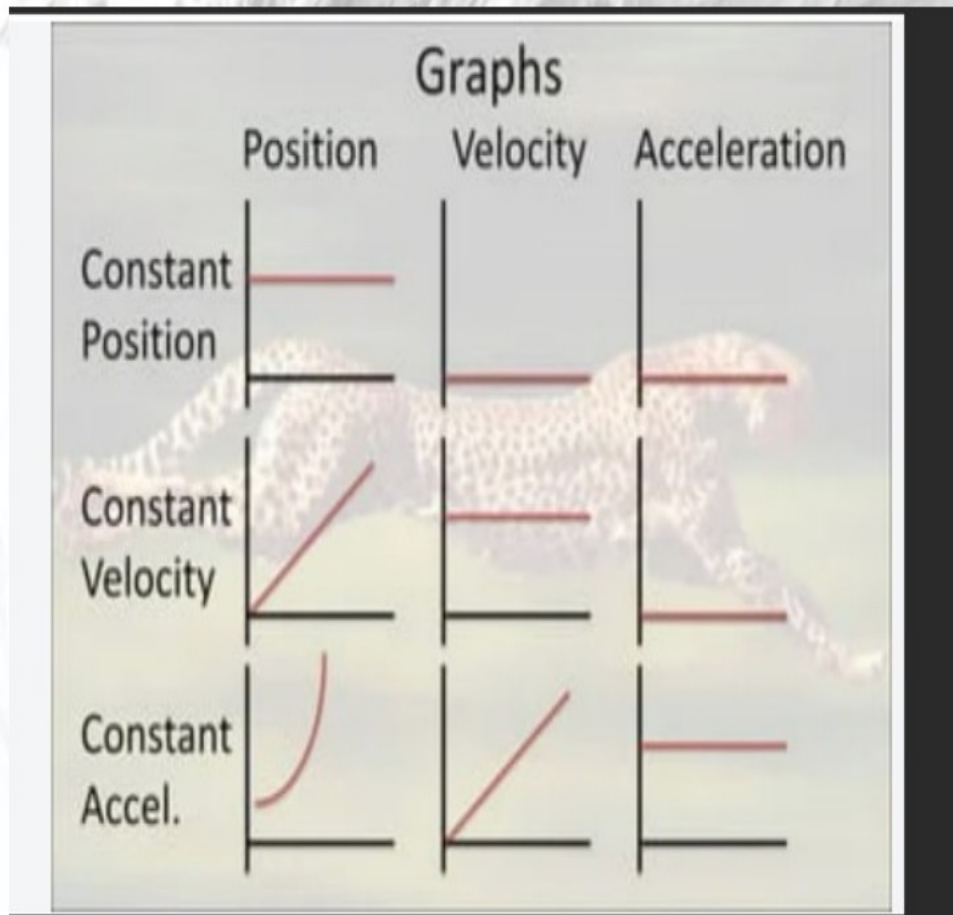
t(s)	x(m)
0	10
1	16
2	24
3	34
4	46

$$a=2\text{m/s/s}$$

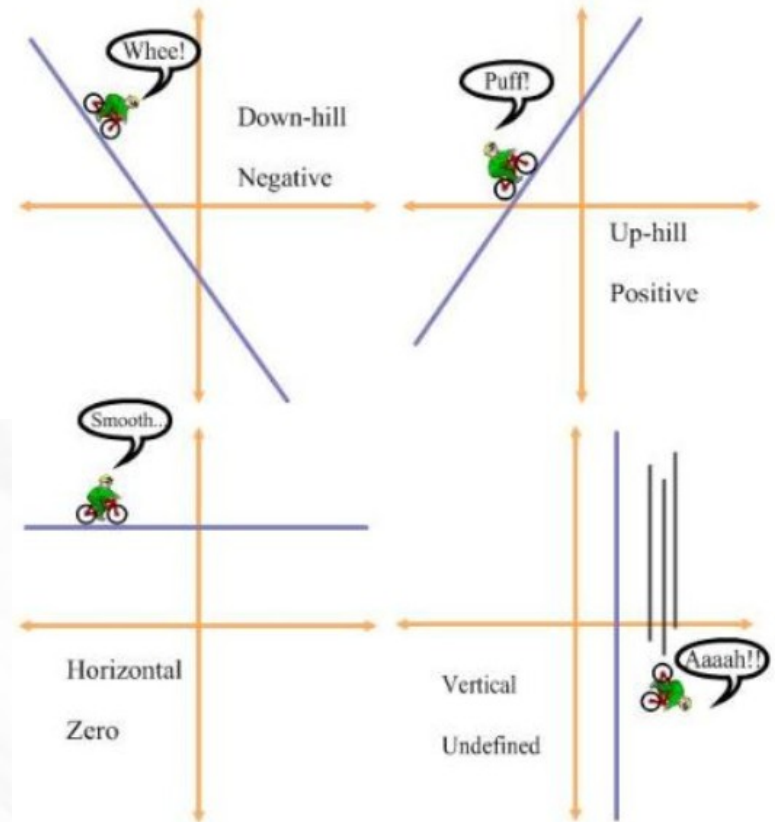
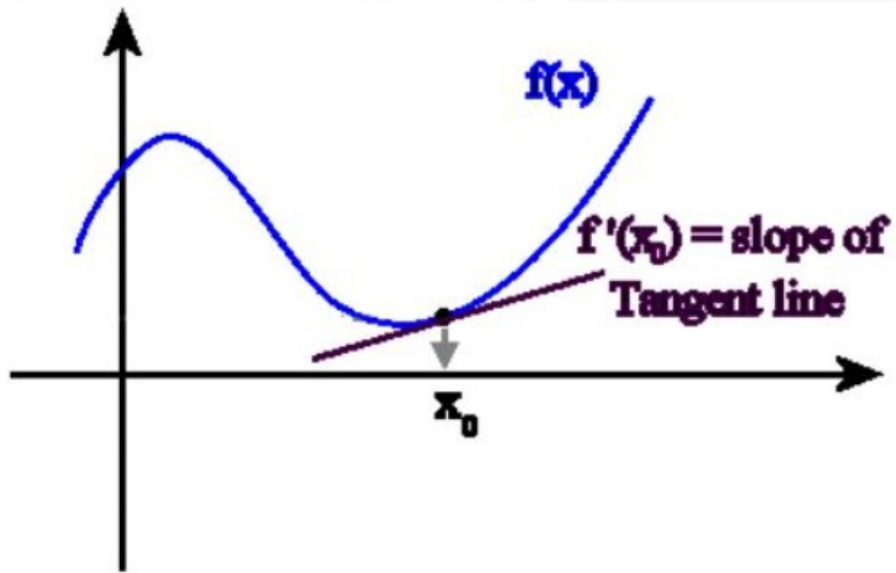
$$V=5 + 2t$$

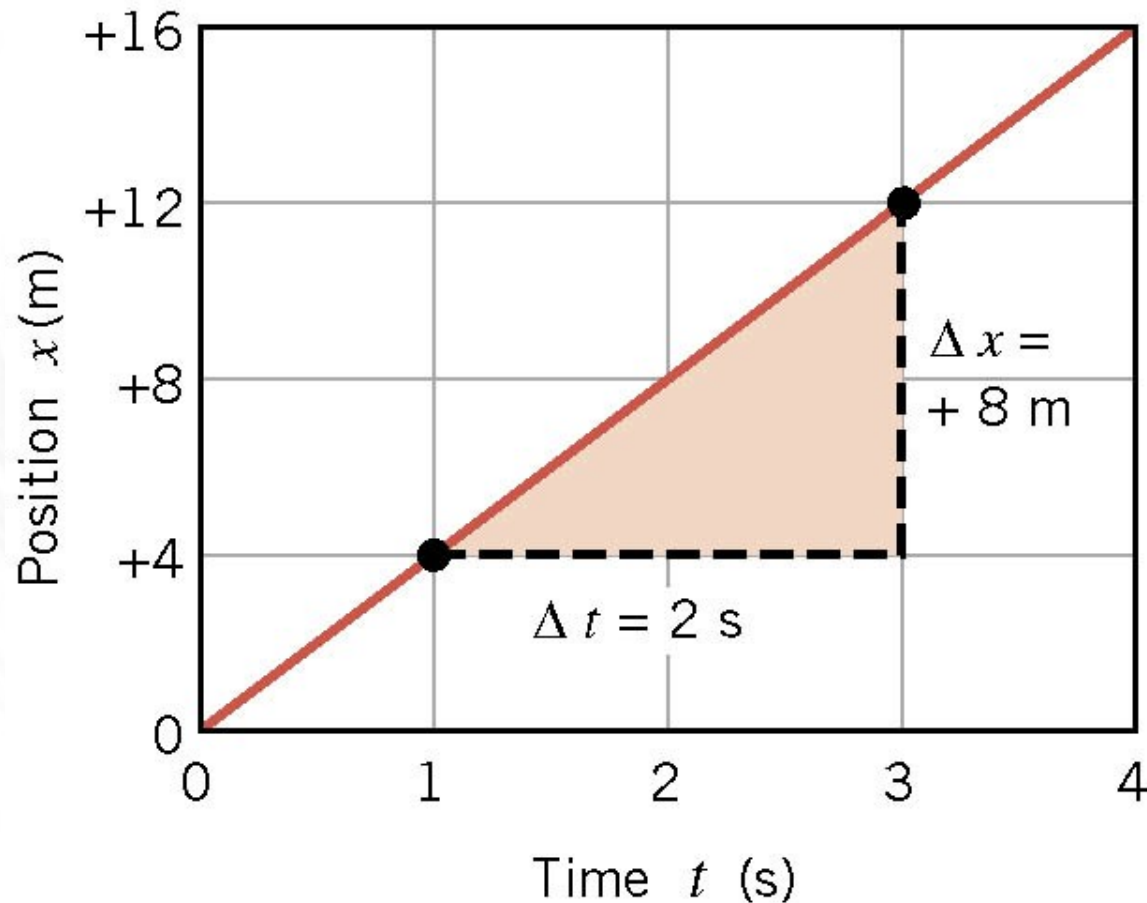
$$X = 10 + 5t + 0.5(2)t^2$$

t(s)	V(m/s)
0	5
1	7
2	9
3	11



Play the gif





$$\text{Slope} = \frac{\Delta x}{\Delta t} = ?$$

What does the slope of the graph position vs time represent

What is the unit ? What is the slope
What is the the equation of motion $x(t)$

Instantaneous Velocity

Instantaneous means “at some given instant”. The instantaneous velocity indicates what is happening at every point of time.

Limiting process:

- Chords approach the tangent as $\Delta t \Rightarrow 0$
- Slope measure rate of change of position

Instantaneous velocity:

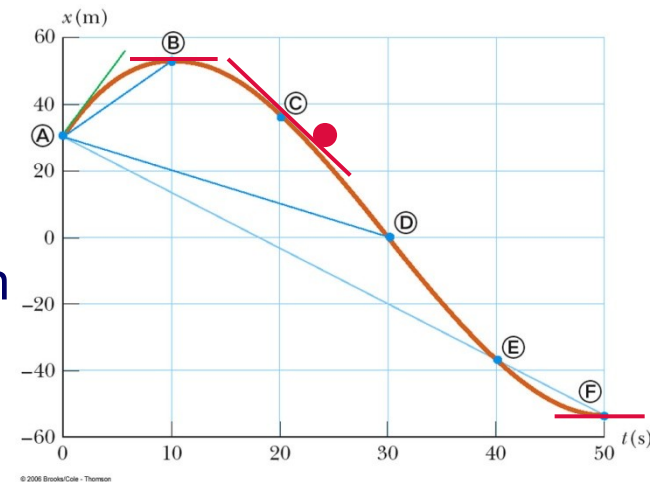
$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

It is a vector quantity.

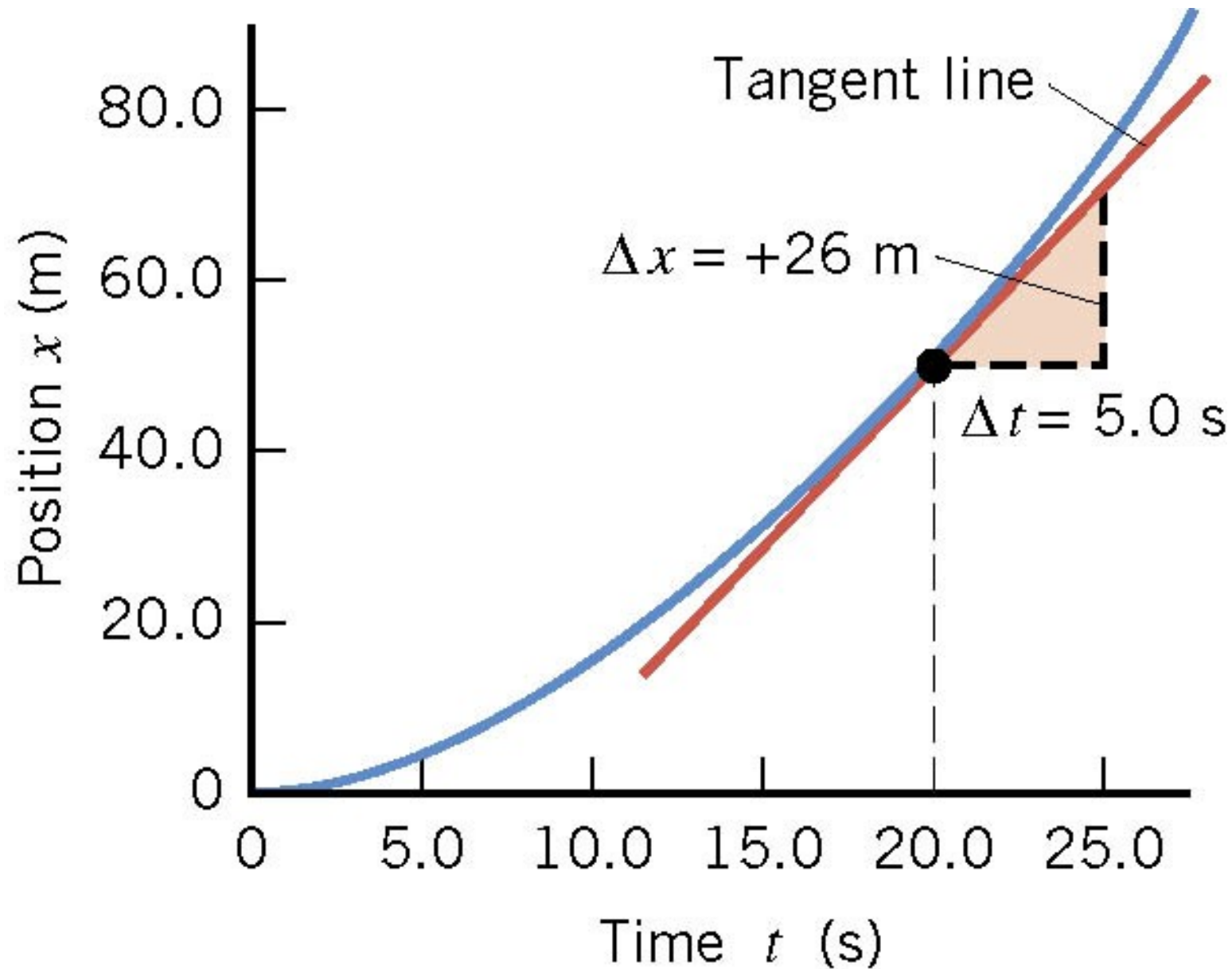
Dimension: length/time (L/T), [m/s].

It is the slope of the tangent line to $x(t)$.

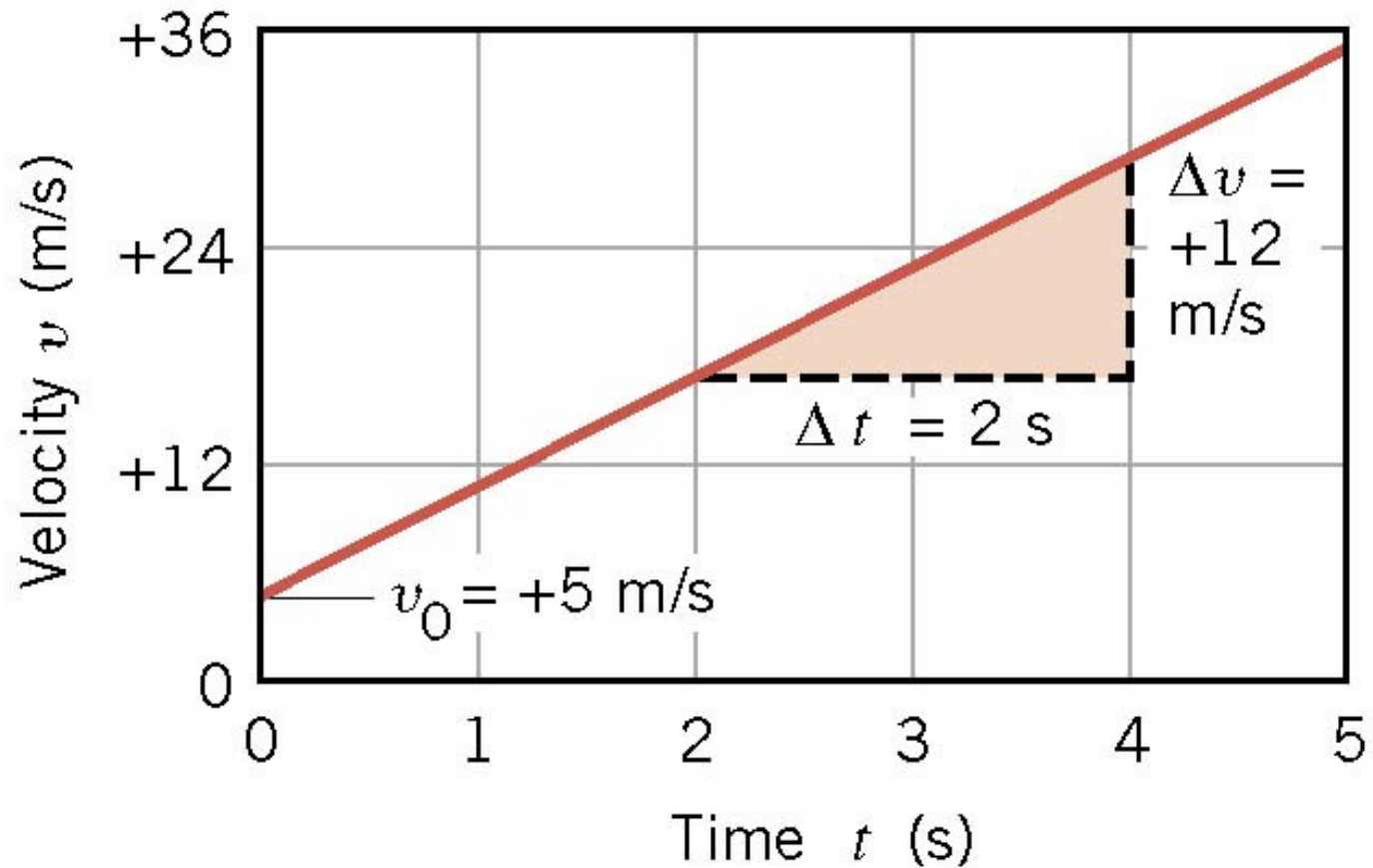
Instantaneous velocity $v(t)$ is a function of time.



2.7 Graphical Analysis of Velocity and Acceleration

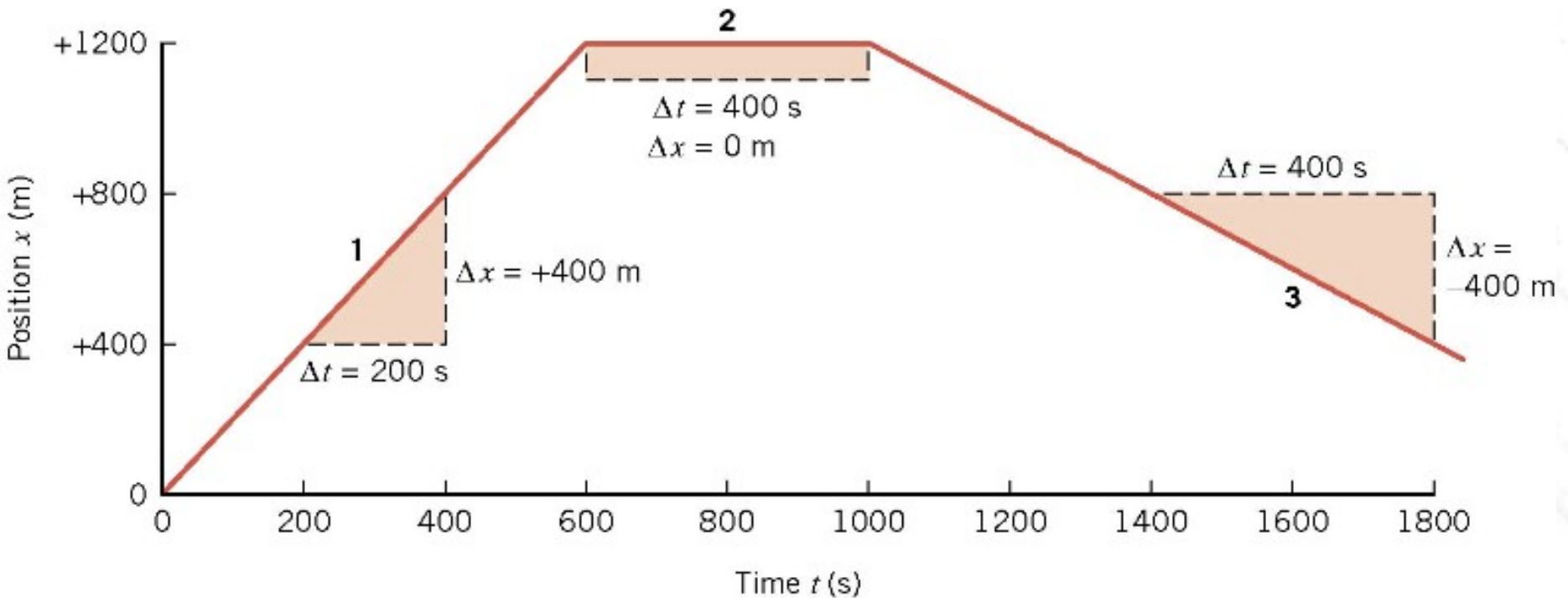
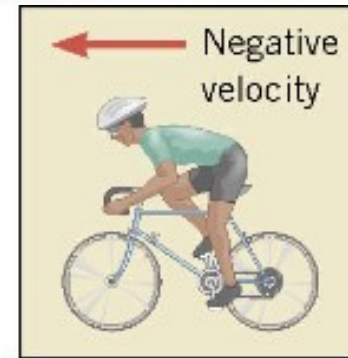
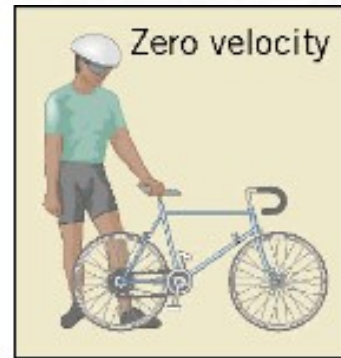
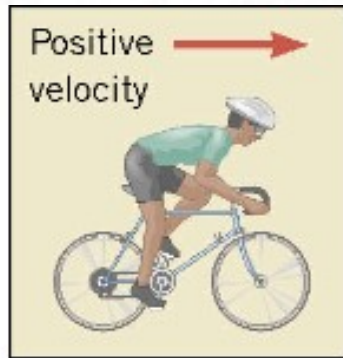


2.7 Graphical Analysis of Velocity and Acceleration

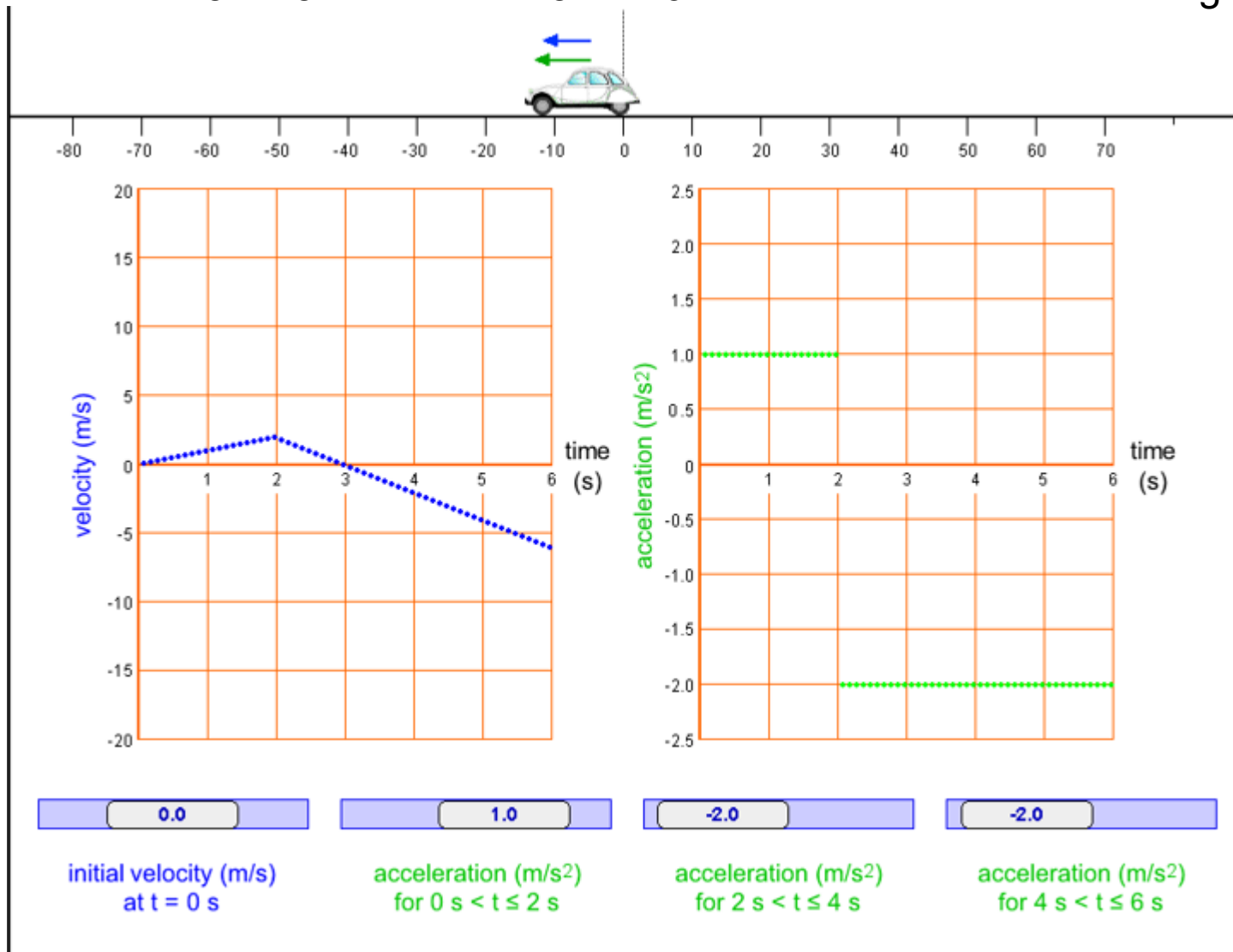


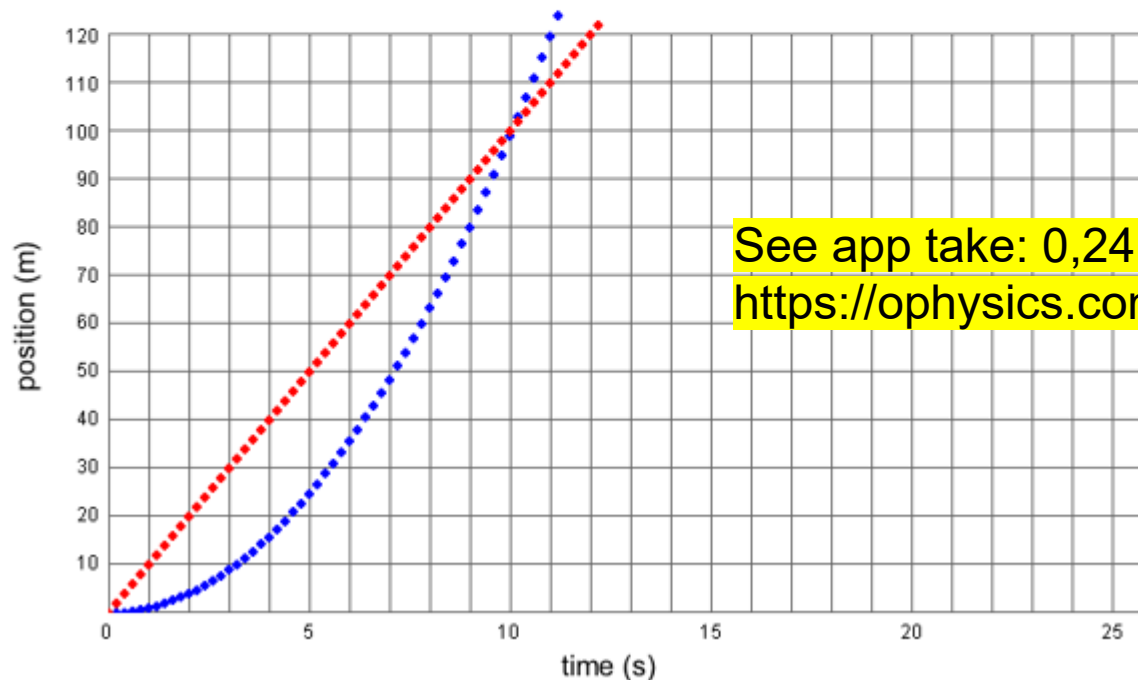
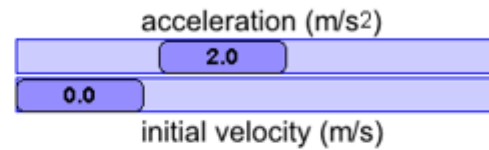
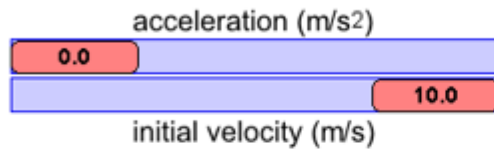
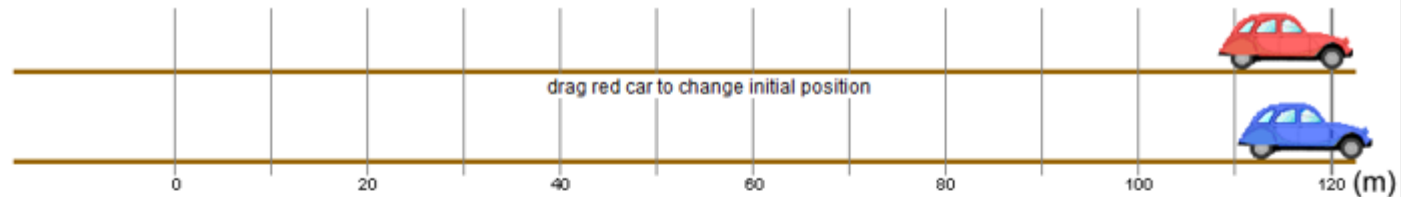
$$\text{Slope} = \frac{\Delta v}{\Delta t} = \frac{+12 \text{ m/s}}{2 \text{ s}} = +6 \text{ m/s}^2$$

2.7 Graphical Analysis of Velocity and Acceleration



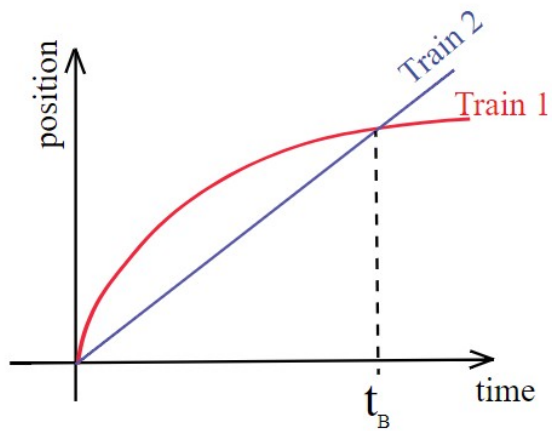
Try to recreate the motion of the car. What's happening between 0 and 2 s? Between 2 and 3 s? 3 s? Between 3 and 6 s? Draw the car for each segment.





See app take: 0,24,0,0,0,10
<https://ophysics.com/k7.html>

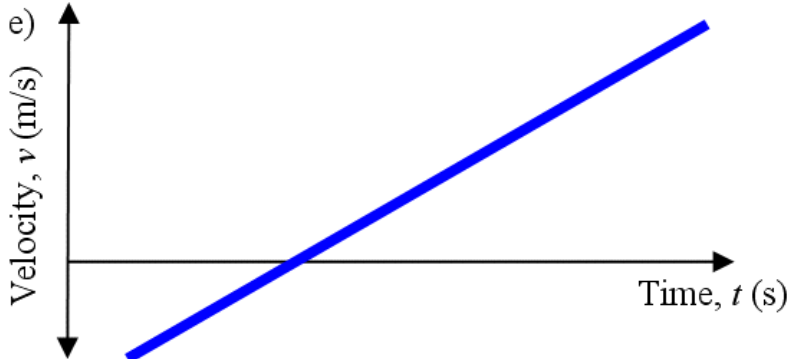
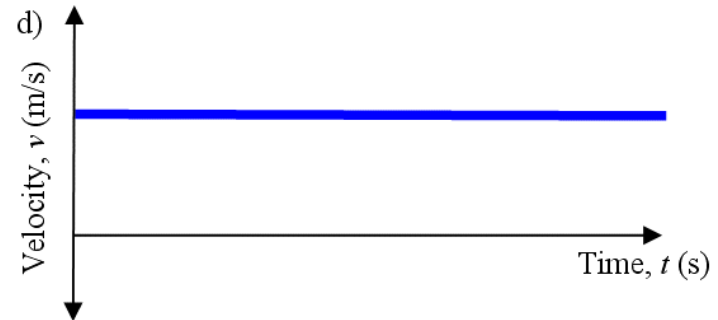
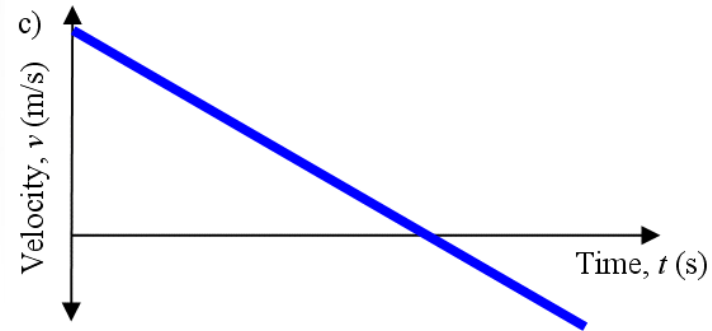
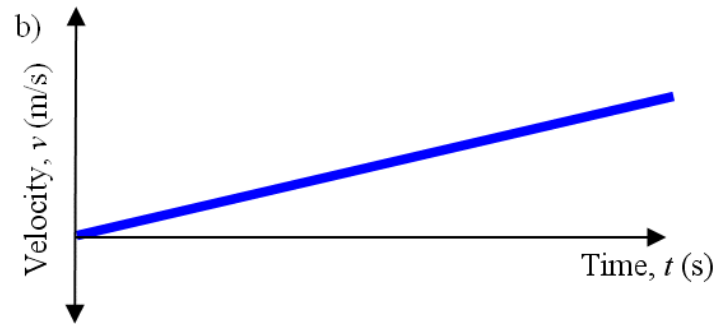
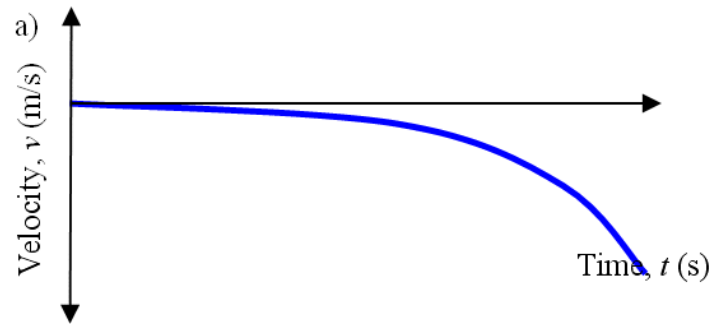
Textbook 2.5 (university physics)



Here are the positions of 2 trains.

- 1) describe the motion of train1 and train2
(constant vel ? Speeding up ? Slowing down ?)
- 2) What happens at t_B ?
- 3) Can you see when the 2 trains have the same velocity ?

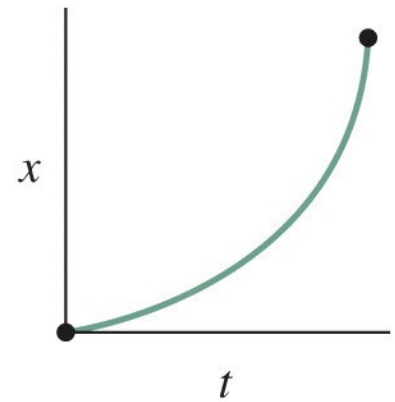
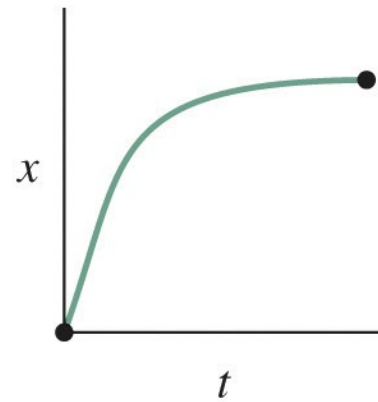
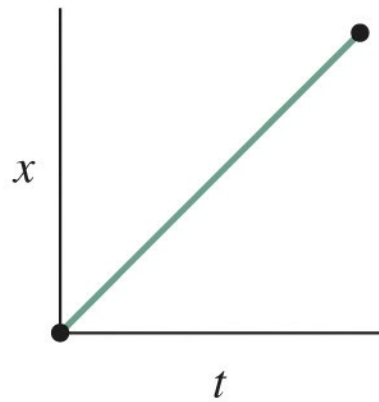
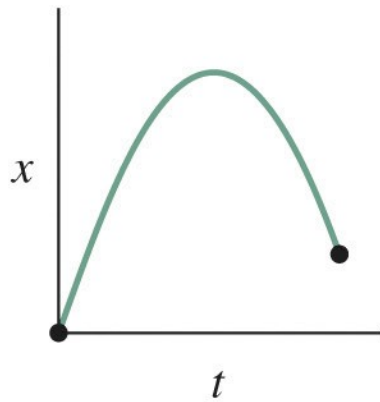
. Which of the following velocity vs. time graphs represents an object with a negative constant acceleration?



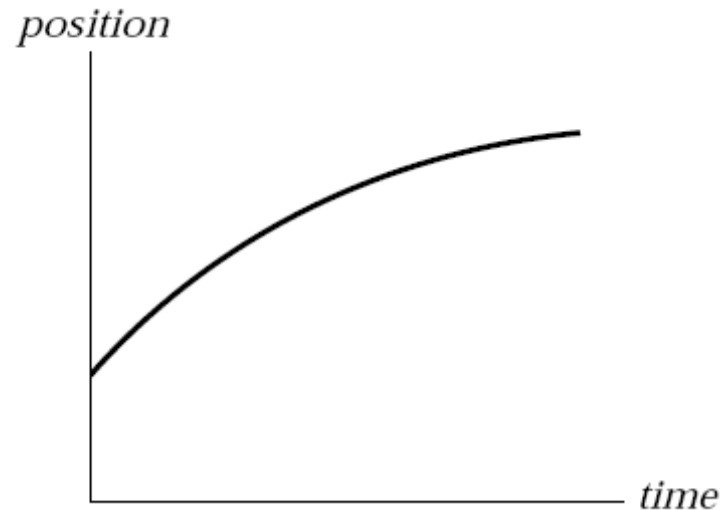
Clicker Question 1



- The figure shows position-versus-time graphs for four objects. Which starts slowly and then speeds up?



A train car moves along a long straight track. The graph shows the position as a function of time for this train. The graph shows that the train:



1. speeds up all the time.
2. slows down all the time.
3. speeds up part of the time and slows down part of the time.
4. moves at a constant velocity.

Clicker Question 2



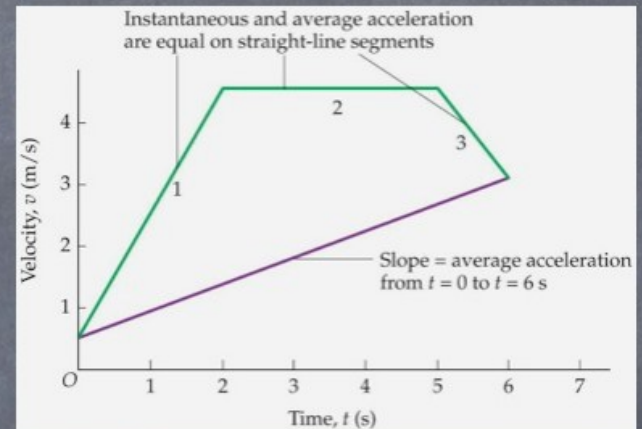
- A speeding motorist zooms past a stationary police car, which then heads after the speeder. The police car starts with zero velocity and is going at twice the car's velocity when it catches up to the car. So at some intermediate instant the police car must be going at the same velocity as the speeding car. When is that instant?
 - A. Closer to the time when the police car starts chasing
 - B. Closer to the time when the police car catches the speeder
 - C. Halfway between the times when the two cars coincide

Hint: For a constant acceleration $V = at$

So if velocity is $2V$ then time is $2t$

Acceleration is 10mph/s if $t = 5\text{ s}$ $V = 20$ then $t = 10\text{ s}$ $V = 40$

A train moving in a straight line with an initial velocity of 0.50 m/s accelerates at 2.0 m/s^2 for 2.0 s , coasts with zero acceleration for 3.0 s , and then accelerates at -1.5 m/s^2 for 1.0 s .

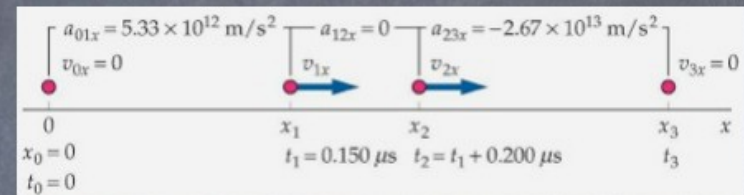


- (a) What is the final velocity v_f of the train?
- (b) What is the average acceleration a_{av} of the train?

Try it on your own.

An electron in a cathode-ray tube accelerates from rest with a constant acceleration of $5.33 \times 10^{12} \text{ m/s}^2$ for $0.150 \mu\text{s}$, then drifts with a constant velocity for $0.200 \mu\text{s}$, then slows to a stop with a negative acceleration of $-2.67 \times 10^{13} \text{ m/s}^2$. (Note: $1 \mu\text{s} = 10^{-6} \text{ s}$).

How far does the electron travel?



1) Draw the electron (as a dot).

Calculate the displacement Δx_i and velocity v_i for each part of the path:

3. The answers have the correct units and appear to be reasonable.

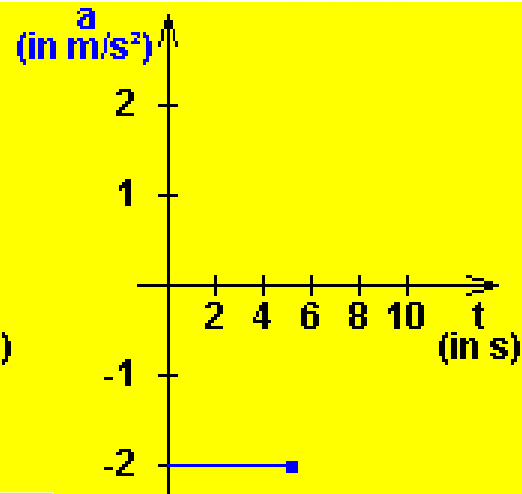
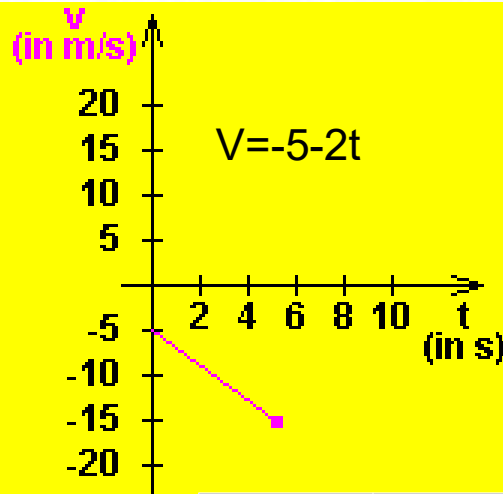
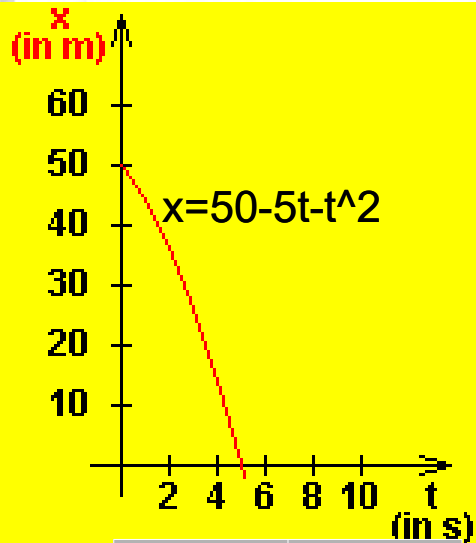
Speeding up @ down or speeding up @ left



$t=0$

Velocity and acceleration have the same direction → speeding up

Example: dropping a ball on a planet with gravity = -2m/s/s = acceleration.

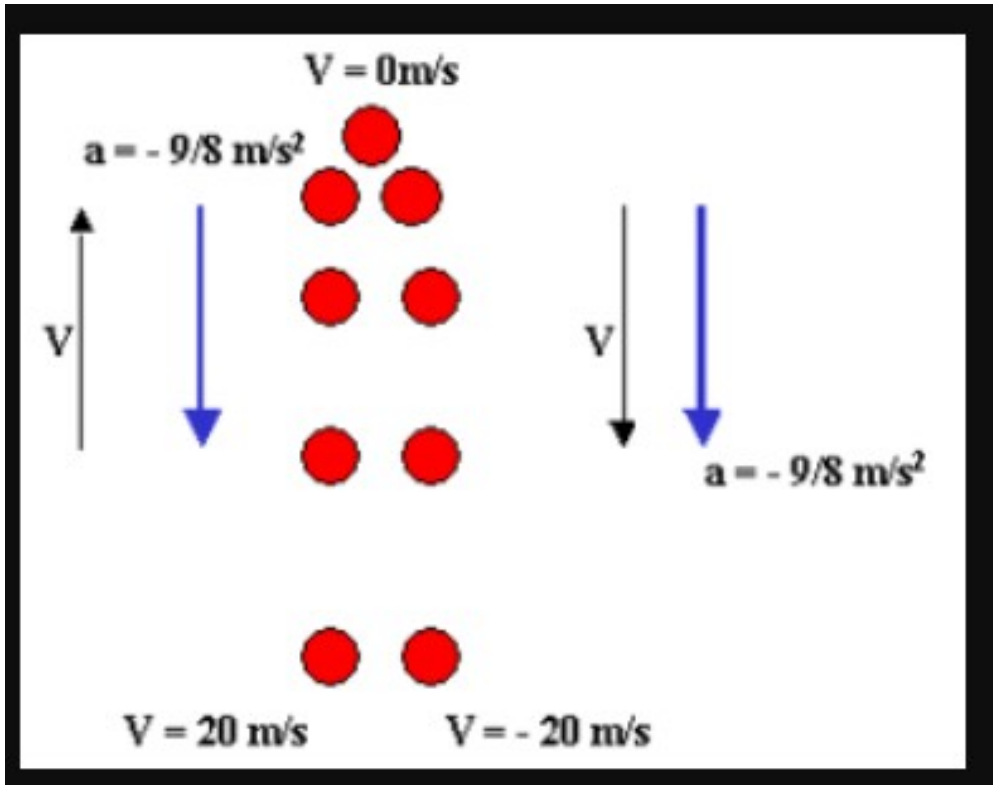


t	v
0	-5
1	-7
2	-9
3	-11

t	x
0	50
1	44
2	36
3	26



Similar situation:



Throwing a ball straight up.

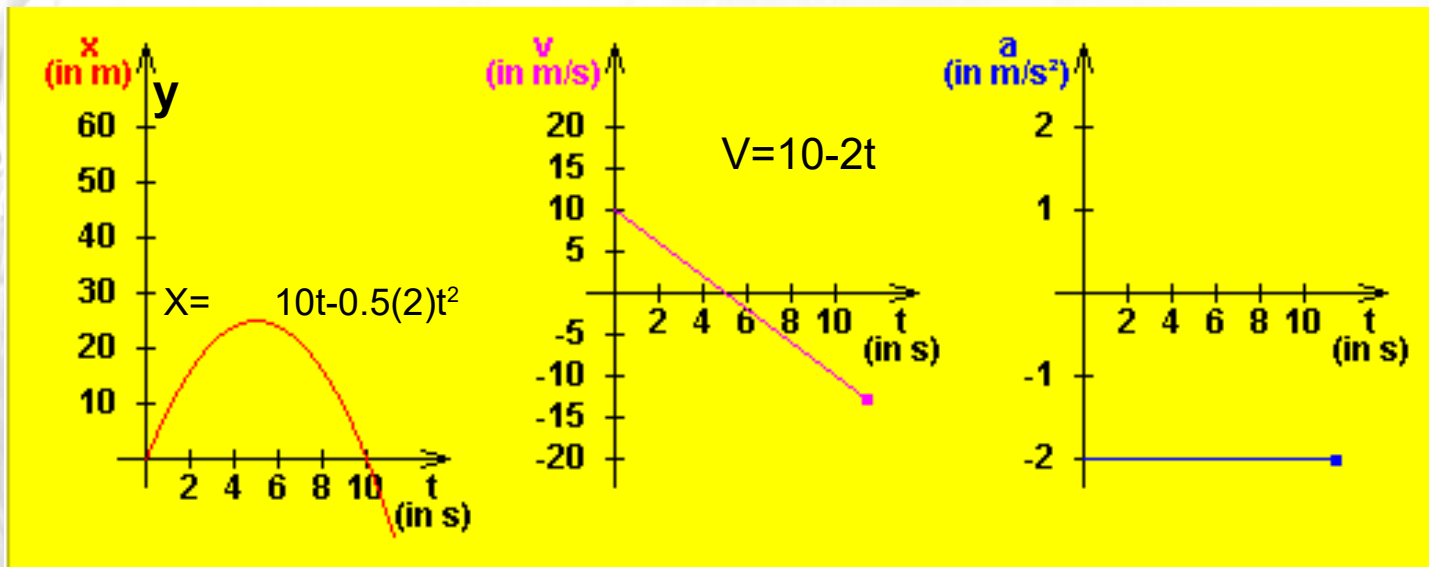
The weight (force) is always pulling @ down. So acceleration is always @ down (negative)

On the way up the force causes The ball to slow down.

On the way down the force causes The ball to speed up.

Trace the graphs for a , V and y

Consider the motion of an object thrown in the air with an initial velocity V_0 . It moves @ up but slows down, stops and changes direction and moves @ down but speeds up. $y(t)$ is the position of the object.
(This is not happening on Earth. But on another viable planet)



t	x
0	0
1	9
2	16
3	21
4	24
5	25
6	24
7	21
8	16
9	9
10	0

82

t	V
0	10
1	8
2	6
3	4
4	2



Clicker Question 7



- If a car has non-zero acceleration, which of the following quantities must be changing?
 - A. Velocity
 - B. Direction
 - C. Speed
 - D. All of the above
 - E. Both direction and speed

Play with: <https://ophysics.com/k4.html>

The 3 minutes calculus

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

Position $\xrightarrow{\frac{d}{dt}}$ Velocity $\xrightarrow{\frac{d}{dt}}$ Acceleration

Graphical relationship between

x , v , and a

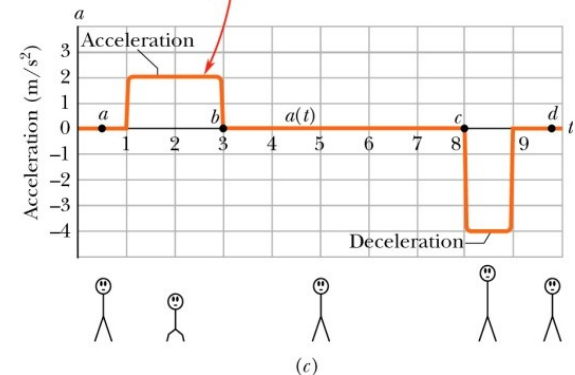
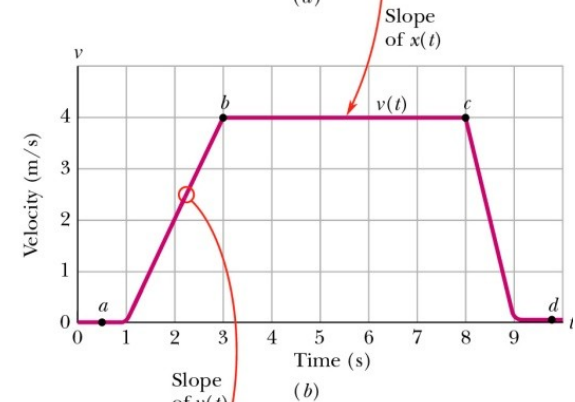
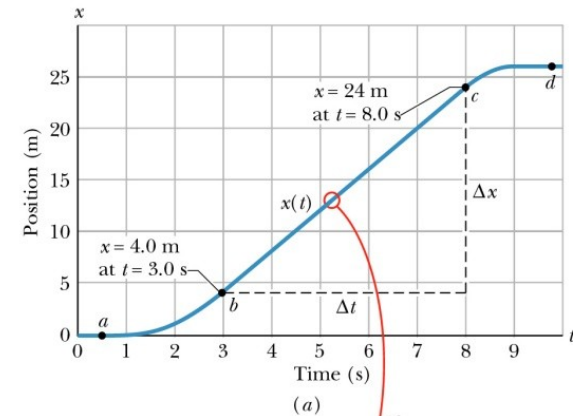
This same plot can apply to an elevator that is initially stationary, then moves upward, and then stops. Plot v and a as a function of time.

Graph $x(t) \rightarrow$

Straight line = velocity is constant $\rightarrow a=0$

Curving up = slope increases $\rightarrow$ velocity increases if moving @ right $\rightarrow a \neq 0$

Curving down = slope decreases if moving @ right $\rightarrow$ velocity decreases $\rightarrow a \neq 0$



We suppose the object is moving @ right

An object starts from rest at time $t = 0.00$ s and moves in the $+x$ direction with constant acceleration. The object travels 12.0 m from time $t = 1.00$ s to time $t = 2.00$ s. What is the acceleration of the object?

A jet plane is launched from a catapult on an aircraft carrier. In 2.0 s it reaches a speed of 42 m/s at the end of the catapult. Assuming the acceleration is constant, how far did it travel during those 2.0 s?

A car starts from rest and accelerates with a constant acceleration of 1.00 m/s^2 for 3.00 s. The car continues for 5.00 s at constant velocity. How far has the car traveled from its starting point?

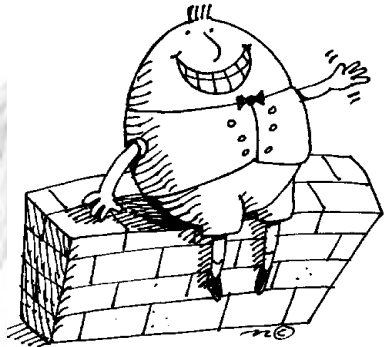
Car A is traveling at 22.0 m/s and car B at 29.0 m/s. Car A is 300 m behind car B when the driver of car A accelerates his car with a uniform forward acceleration of 2.40 m/s^2 . How long after car A begins to accelerate does it take car A to overtake car B?

More
challenging

) A car is traveling at 26.0 m/s when the driver suddenly applies the brakes, causing the car to slow down with constant acceleration. The car comes to a stop in a distance of 120 m. How fast was the car moving when it was 60.0 m past the point where the brakes were applied?

Free-Fall

Kinematics equation in 1 D along vertical



Reference: The physics of everyday phenomena: a conceptual introduction to physics
by W. Thomas Griffith, McGraw-Hill

Conceptual Physics – Paul Hewitt

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2.6 *Freely Falling Bodies*

In the absence of air resistance, it is found that all bodies at the same location above the Earth fall vertically with the same acceleration. If the distance of the fall is small compared to the radius of the Earth, then the acceleration remains essentially constant throughout the descent.

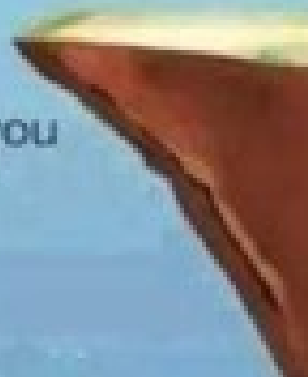
This idealized motion is called *free-fall* and the acceleration of a freely falling body is called the *acceleration due to gravity*.

$$g = 9.80 \text{ m/s}^2 \quad \text{or} \quad 32.2 \text{ ft/s}^2$$

First Law of Cartoon Physics:

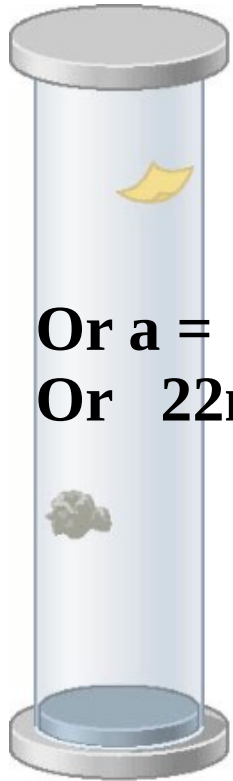


"Gravity doesn't work until you
look down"



Free-Fall = no air resistance

All objects, regardless of their masses, have the same acceleration



Or $a = 10\text{m/s/s}$

Or 22mph/s or 32ft/s/s

Air-filled
tube
(a)



Evacuated
tube
(b)

$a=g=-10\text{m/s/s}$ (more precisely -9.8m/s/s)

We are going to use the kinematics equations with

The object is acted upon by only the weight.

Weight is pulling down, this is why acceleration is @ down. We neglect air resistance.

Do demo vacuum tube with penny
And feather. +book and paper +

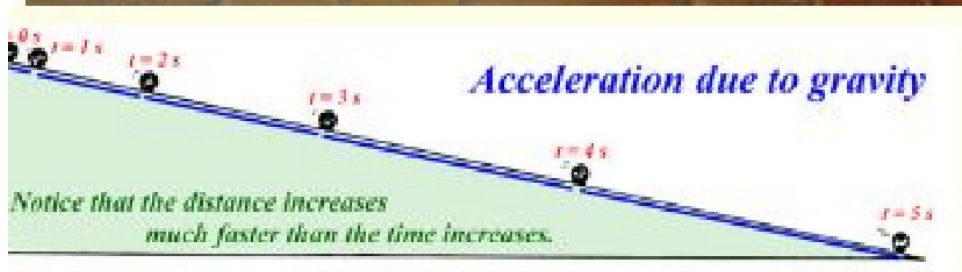
Video Moon and demo



Odd number rule. The tower of Pisa is a myth

http://dev.physicslab.org/Document.aspx?doctype=2&filename=Kinematics_GalileoRamp

<https://catalogue.museogalileo.it/multimedia/InclinedPlane.html>



How to use the equations of kinematics free-fall:

$a = -10\text{m/s/s}$ and $Y_0 = 0$

Y is the vertical position in on the Y axis (y -coordinate)

$$\Delta Y = V_0 t - 0.5(10) t^2$$

$$\text{Or } \Delta Y = V_0 t - 5 t^2$$

$$V = V_0 - 10t$$

$$V^2 = V_0^2 + 2(-10)(Y)$$

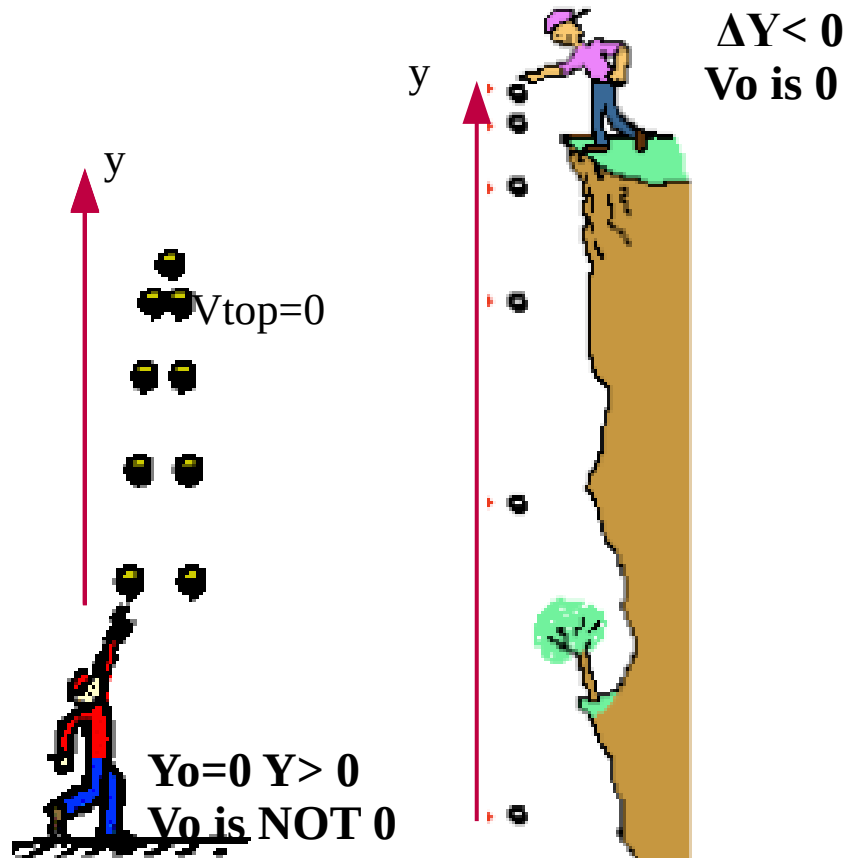
For now are are going to work with 2 cases.

See how the Y axis defines the sign of the position Y .

In both cases $Y_0 = 0$

But the initial velocity is $V_0 = 0$ in one case

And not 0 in another case.



Source images:

Paul Hewitt

Conceptual Physics

$$\Delta Y = V_0 t - 5 t^2$$

$$V = V_0 - 10t$$

$$V^2 = V_0^2 + 2(-10)(\Delta Y)$$

$$V_{\text{top}} = 0$$

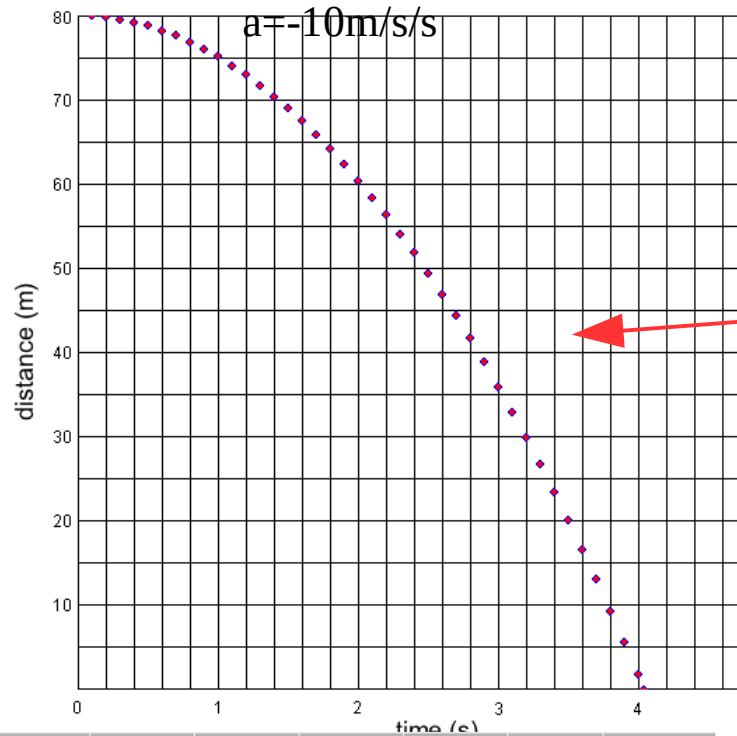
$$\Delta Y = -5 t^2 \text{ or } Y = Y_0 - 5 t^2$$

$$V = -10t$$

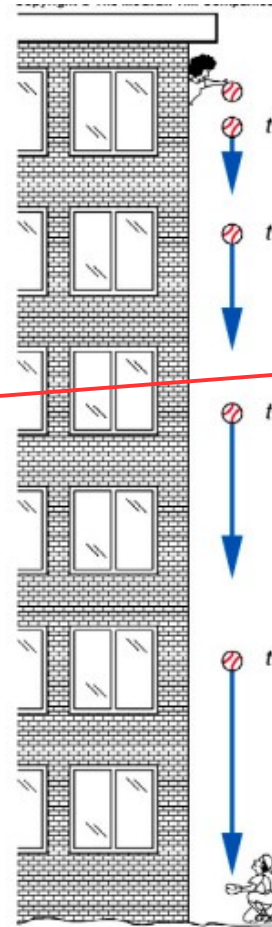
$$V^2 = 2(-10)(\Delta Y)$$

$$V_0 = 0$$

See exploration physical sciences



t	0	1	2	3	4	5	
V	0						
Δy	0						
Y	80						

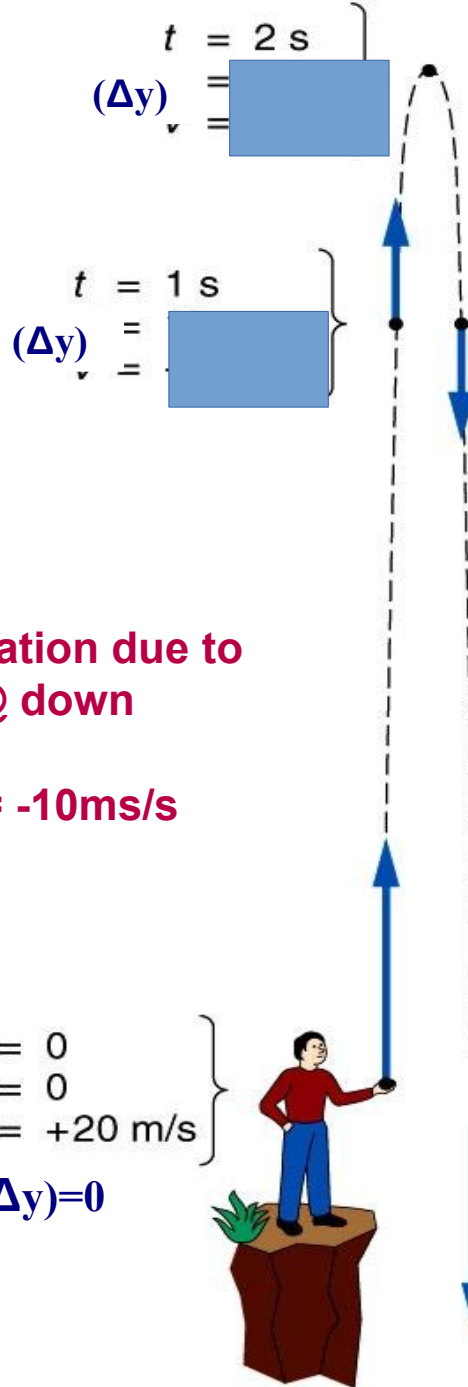


See exploration of physical science Applet # 3

Applet # 3 observe the graph.
The slope is negative or positive ?
Is the speed increases or decreases ?
(think steepness)

Now add air resistance.
What is happening to the graph after
A few second ?? The objects reaches
Terminal speed.

Complete the table. Kinematics equations:
Use $V_0 = 0$ and $V = -10t$
 $\Delta Y = -5 t^2$ with $a = -10$ and $V_0 = 0$ and $y_0 = 80$



REMEMBER : the weight never stops pulling
 So the acceleration is always -10 m/s^2 even at
 The top ! (for falling body)
 Acceleration = weight per unit mass

The acceleration due to Gravity is @ down

Suppose $g = -10 \text{ m/s}^2$

Equations kinematics with $a = -10 \text{ m/s}^2$
 y is the vertical position. Here $V_0 = 20 \text{ m/s}$

$Y_0 = 0$ with $\Delta y = 20t - 5t^2$

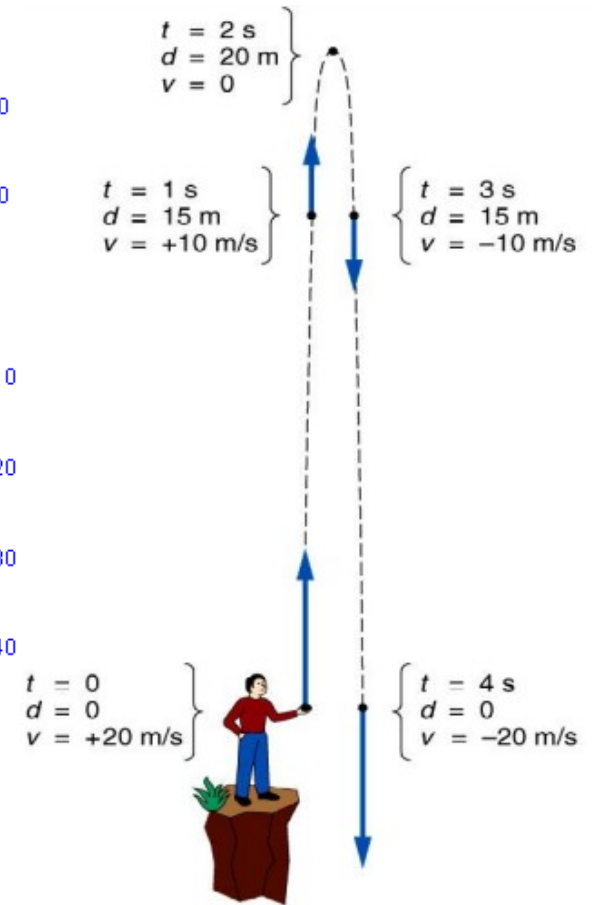
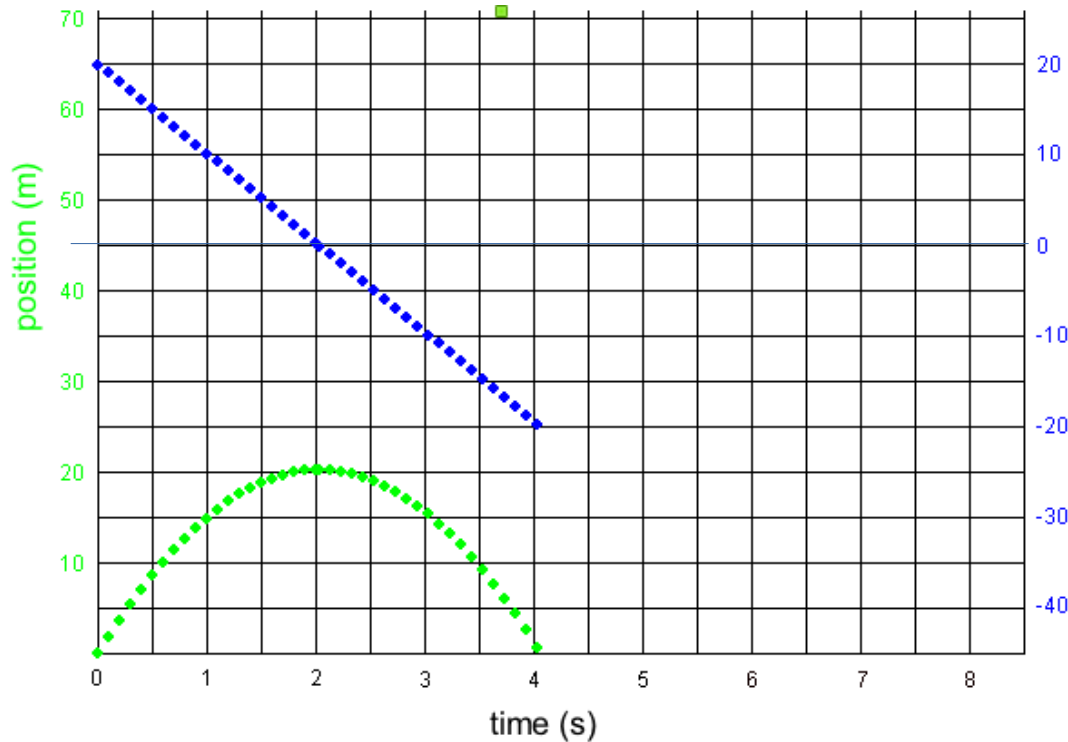
(initial position $y_0 = 0$ and $d = y$ here)

$V = 20 - 10t$

$V^2 = 20^2 + 2(-10)(\Delta y)$

t	0	1	2	3	4		
V							
Δy							

Applet 5: important $V = 0$ at the top of the trajectory !



Time it takes to go up = time it takes to do down

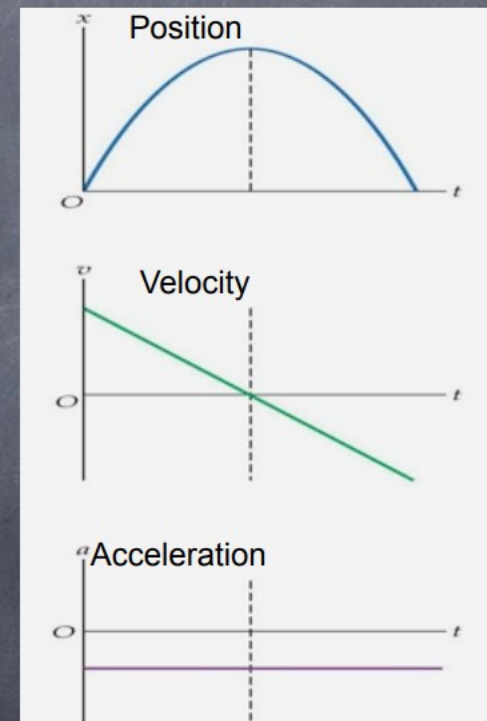
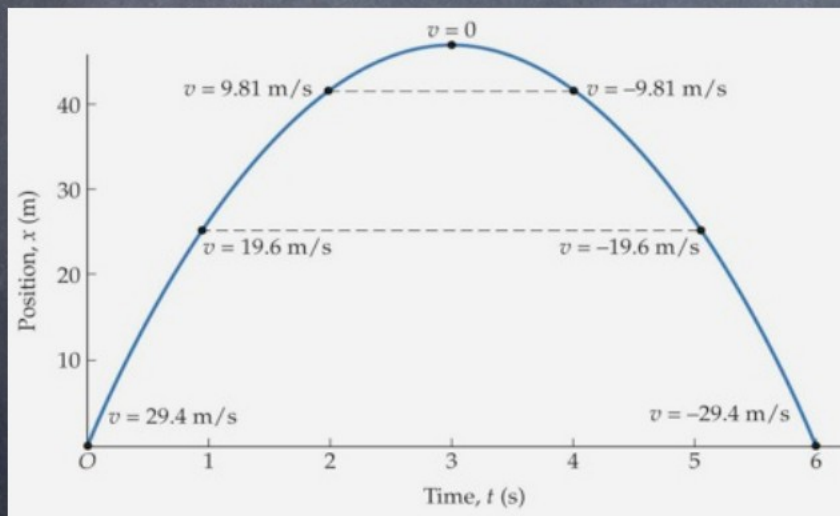
Time of flight = 2 x time it takes to go up

At the top $V(t=2) = 0$ but $a = -10 \text{ m/s/s}$ at all time.

• The motion may be symmetrical

• then $t_{\text{up}} = t_{\text{down}}$

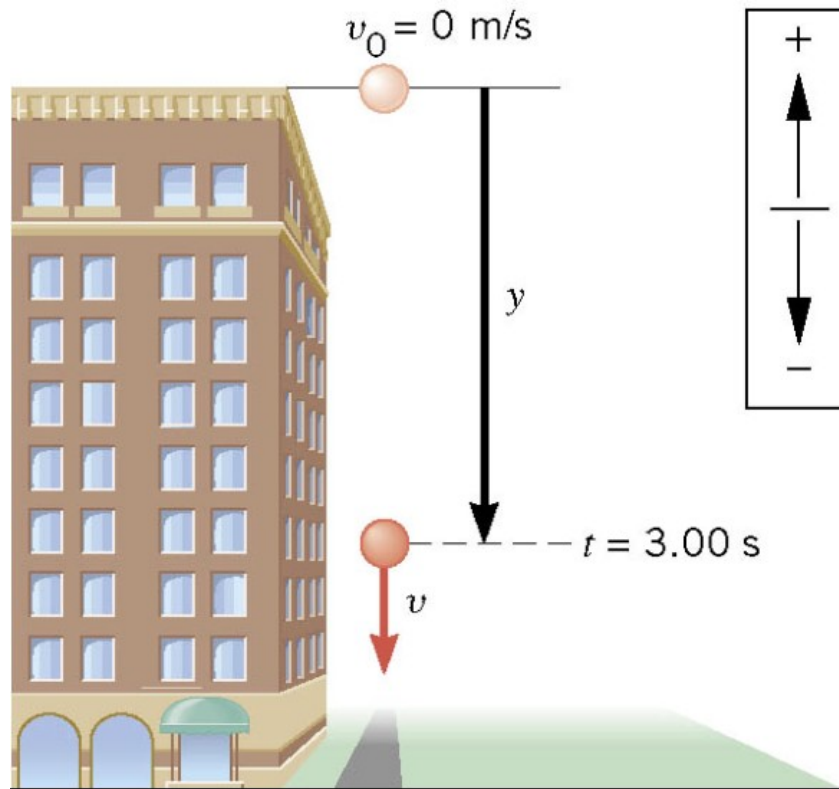
• then $v_f = -v_o$



2.6 Freely Falling Bodies

Example 10 A Falling Stone

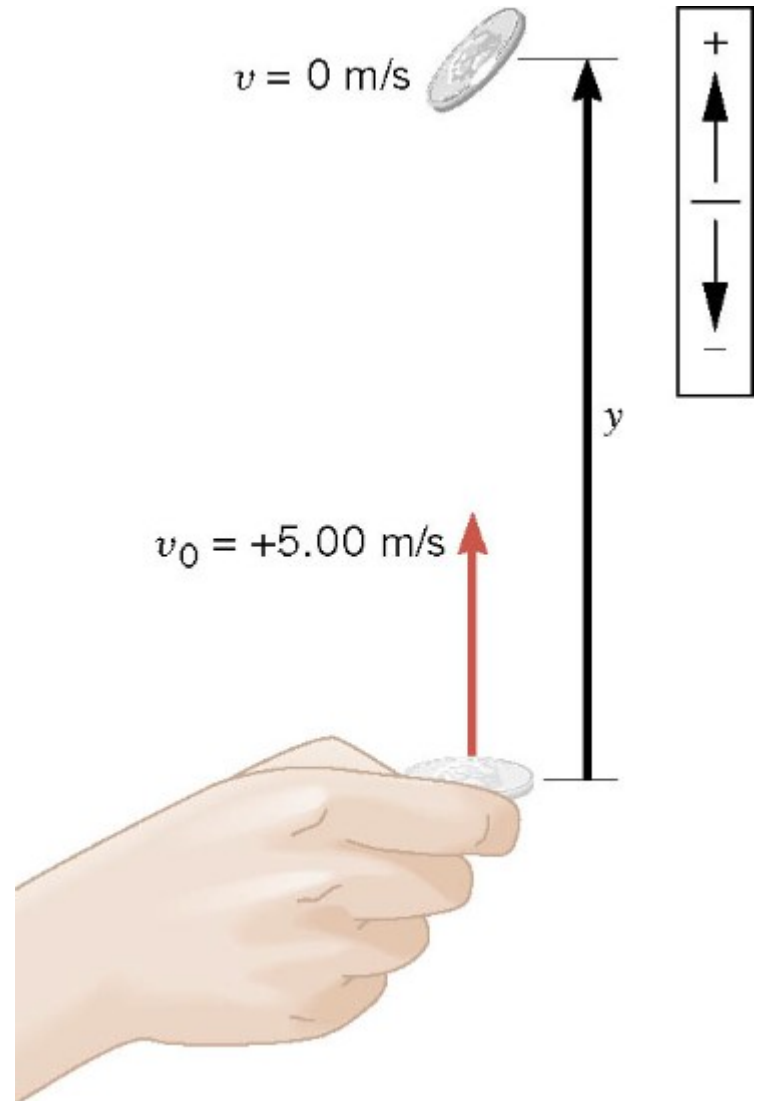
A stone is dropped from the top of a tall building. After 3.00s of free fall, what is the displacement y of the stone?



2.6 Freely Falling Bodies

Example 12 How High Does it Go?

The referee tosses the coin up with an initial speed of 5.00m/s. In the absence of air resistance, how high does the coin go above its point of release?



2.6 *Freely Falling Bodies*

Conceptual Example 14 Acceleration Versus Velocity

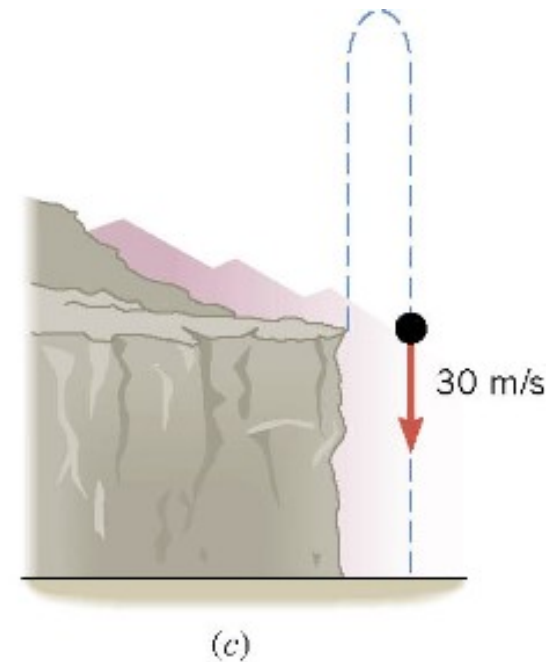
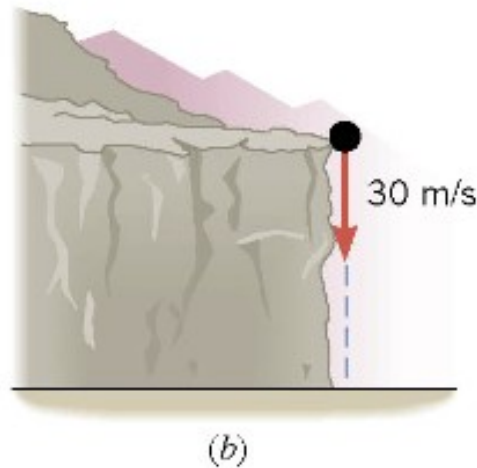
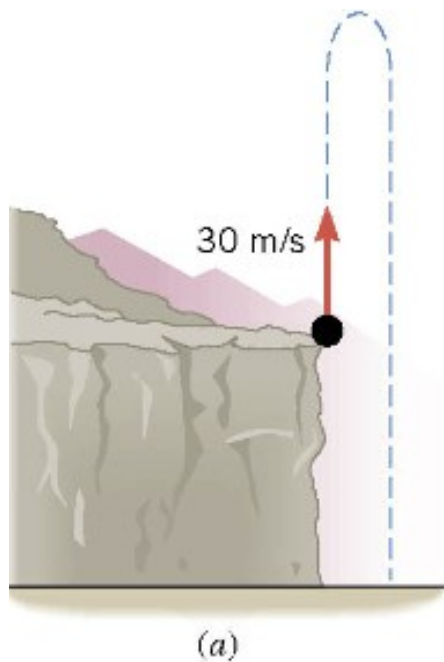
There are three parts to the motion of the coin. On the way up, the coin has a vector velocity that is directed upward and has decreasing magnitude. At the top of its path, the coin momentarily has zero velocity. On the way down, the coin has downward-pointing velocity with an increasing magnitude.

In the absence of air resistance, does the acceleration of the coin, like the velocity, change from one part to another?

2.6 Freely Falling Bodies

Conceptual Example 15 Taking Advantage of Symmetry

Does the pellet in part *b* strike the ground beneath the cliff with a smaller, greater, or the same speed as the pellet in part *a*?



Clicker Question 4



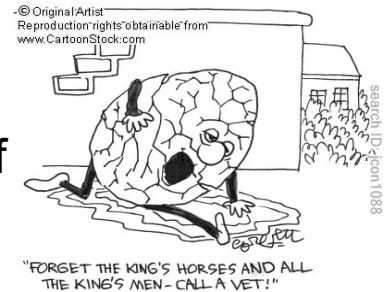
- Standing on a roof, you simultaneously throw one ball straight up and another one down with the same initial speed. Neglecting air resistance, which ball hits the ground with greater speed?
- A. The ball dropped from rest
- B. The ball thrown straight up
- C. Both balls have the same speed.

Make a graph to solve !!!

Example 2.7 and 2.9 textbook

Use TI to graph

- 1) a high wire artist missteps and falls 9.2m to the ground. What is her speed on landing ?
Convert to mph.
- 2) A cat falls out of a tree, dropping 16m to the ground. How long is the cat in the air ?
- 3) You drop a rock from a bridge, and it hits the water 2.3 s later. Find:
A) the height of the bridge
B) the velocity of the rock when it hits.
- 4) A steel ball is dropped from a diving platform (with an initial velocity of zero). Using $g = -10\text{m/s}^2$
A) what is the velocity of the ball 0.8s after its release
B) What is its velocity after 1.6s after its release?
C) through what distance does the ball fall in the first 0.8s seconds of
D) How far does it fall in the first 1.6s of its flight ?
- 5) and bowling ball and a tennis ball are released from rest at the same
Time from a height of 50m.
A) Which one will reach the ground first
B) How long it will take ?



6) A ball is thrown downward with an initial velocity of 12m/s (initial = -12) Using $g = -10\text{m/s/s}$, what is the velocity of the ball 1 second After it is released ?

7) A ball is dropped from a high building. Using the approximation $g = -10\text{m/s/s}$, find the **change** in velocity between the first and the fourth Second of its flight.

8) A ball is thrown upward with an initial velocity of 15m/s/s . Using $g = -10\text{m/s/s}$.

What are the magnitude and the direction of the

Ball's velocity. A) 1 second after it is thrown

B) after 2 seconds after it is thrown

C) how high above the ground is the ball after 1 second after it is thrown ?

D) after 2 seconds

E) At what time the ball reach the high point in its flight.

9) Suppose that the gravitational acceleration on a certain planet is only 2m/s/s /

A space explorer standing on this planet thrown a ball Straight up

Straight up ward with an initial velocity of 18m/s .

A) What is the velocity of the ball 4s after it is thrown ?

B) How much time elapses before the ball reached the high point in its flight?

C) How long does it stay in the air/

10) a stone is thrown in the air with initial velocity of 20m/s .

How high does it get ?

Time it takes to reach this max ?

Total time of flight ?